

安全排放量設計

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網 要

1. 安全排放系統之目的及步驟。
2. 相關名詞解釋。
3. 安全排放系統之相關法規與標準。
4. 安全排放設備之介紹。
5. 安全排放量之計算。
6. 安全排放設備規格書之開立。
7. 安全排放系統配管應注意事項。

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- 避免人員傷亡。
- 保護設備防止損壞。
- 減少對環境之衝擊。
- 減少財產與保險費用。

1. 判定可能造成過壓之各種可能原因。
2. 計算在各個情況下之排放量。
3. 選擇適當之疏解裝置。
4. 開立疏解裝置之規格書。
5. 與廠商討論疏解裝置之適當尺寸與型號。
6. 決定疏解裝置之安裝位置。
7. 檢討排放管線之設計，要同時考慮到其他相關疏解裝置及下游設備。

Maximum Allowable Working Pressure (MAWP)

MAWP是指設備在扣除腐蝕裕度及其他的負荷所需的厚度後，依據法規計算該設備所能承受的最大壓力，因此MAWP一定不小於設計壓力。通常從細部設計所得到的數據很多是MAWP 等於設計壓力。

Design Pressure (DP)

設計壓力是使用者依據系統在正常操作情況下之可能最大操作壓力，再加上適當的安全係數所得之設計值。

$$DP \leq MAWP$$

Accumulation

設備在安全閥排放時，可以容許壓力超過設備之MAWP，其數值通常是以MAWP的百分比表示。

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Set Pressure (SP)

是指安全閥入口處在設定排放時之壓力值。

Overpressure

設備在安全閥排放時，可容許壓力超過設備之SP數值，通常以SP的百分比表示。

Back Pressure

在安全閥的出口系統中存在的壓力，此背壓為Superimposed與build-up二種壓力的總合

Build-up Back Pressure

當安全閥打開後，在出口管線因流動所增加之壓力。

Superimposed Back Pressure

在安全閥的出口系統已經存在的靜壓力(Static Pressure)，此壓力來自其他系統可能是固定或變動的。

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Blowdown

安全閥的設定壓力與關閉壓力的差值，以SP的百分比表示。

Opening Pressure

Disk開啟到流體排放形成連續狀態的壓力值。

Closing Pressure

Disk由開啟狀態回復到重新接觸閥座狀態的壓力值。

Marked Burst Pressure

Rupture Disks 之Burst Pressure，廠商在設計溫度下之測試值，會在Disk上面標示，同一批會有相同之測試值。

Simmer

在低於SP的壓力下，可目測或聽到壓縮流體開始由密封面洩漏。

Relief Valve

閥的開度與過壓成正比，主要應用於不可壓縮流體(液體)

Safety Valve

達到設定壓力時閥門會迅速打開，主要應用於可壓縮流體(氣體)。

Safety Relief Valve

同時具有Relief與Safety valve之功能，可應用於液體及氣體。

※其他相關名詞詳細請參考API 521之說明

1. API RP 520 “Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries Part I – Sizing and Selection,” 7th ed. Jan., 2000.
2. API RP 521 “Guide for Pressure-Relieving and Depressuring Systems,” 4th ed. Mar., 1997.
3. API Standard 2000 “Venting Atmospheric and Low-Pressure Storage Tanks – Nonrefrigerated and Refrigerated”, 5th ed., Apr. 1998.
4. ASME Boiler and Pressure Vessel Code Section VIII Division I.
5. NFPA 30 “Flammable and Combustible Liquids Code” 2003 Edition.
6. “鍋爐及壓力容器安全規則”, 85.02.14
7. “高壓氣體勞工安全規則”, 87.06.30
8. “公共危險物品及可燃性高壓氣體設置標準及安全管理辦法”, 95.11.01
9. CNS 9788 “壓力容器(通則)”, 90.06.08
10. CNS 9967 “蒸汽及壓力氣體用彈簧式安全閥”, 72.02.11

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疏解裝置種類之介紹

- ◆ Conventional Pressure Relief Valve
排放性能會明顯受背壓影響
- ◆ Balanced Pressure Relief Valve
排放性能比較不會受背壓影響
- ◆ Pilot-Operated Pressure Relief Valve
主體安全閥會結合一輔助安全閥並受其控制
- ◆ Rupture Disk device
主體包括破裂盤及一組夾具

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- ◆ 安全閥入料管線壓降需小於設定壓力之3%。
- ◆ **Conventional** PRV之背壓必須小於 10% Set Pressure (at 10% Allowable Overpressure)，當背壓高於10% Allowable Overpressure且是固定時，也可以藉著調整 Spring Load來克服之。
- ◆ **Balance** PRV之背壓可以達到50% Set Pressure，但是必須諮詢製造廠商來確認PRV之排放量。
- ◆ **Pilot - Operated** PRV，其Valve Lift則不受背壓之影響。
- ◆ 設計者可以參考**API 526**之規範去選擇**PRV**，對材質及其他相關尺寸與結構都有詳細之規定。
- ◆ **API 527**則是對PRV之**Seat Tightness**有明確之規定。

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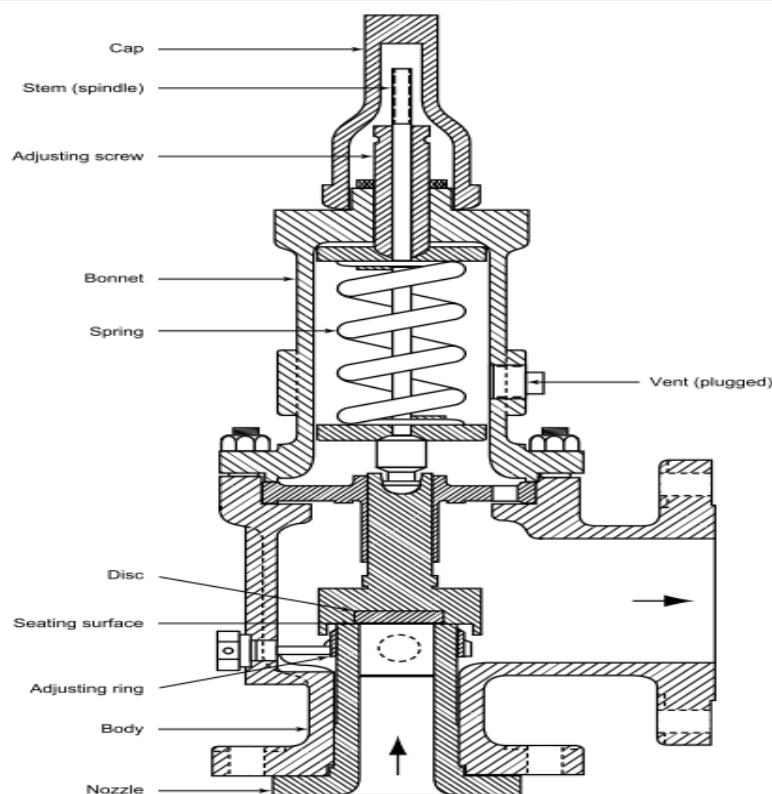


Figure 2—Conventional Pressure Relief Valve with a Single Adjusting Ring for Blowdown Control

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Relationship Between Lift of Disk in PRV & Vessel Pressure

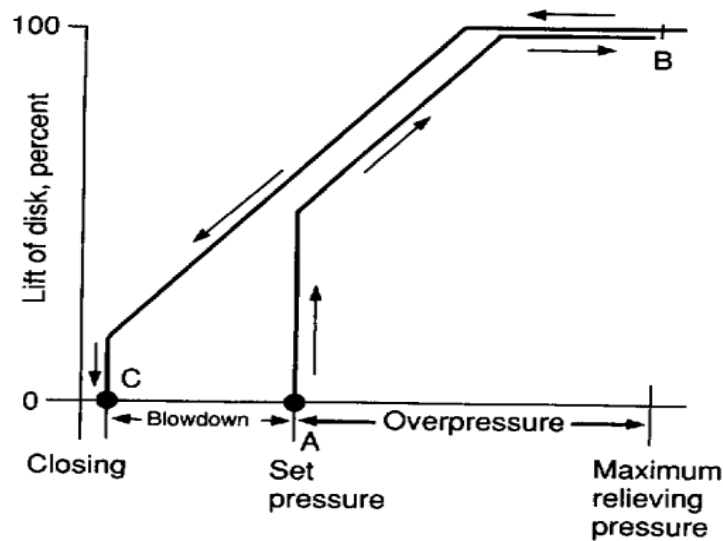
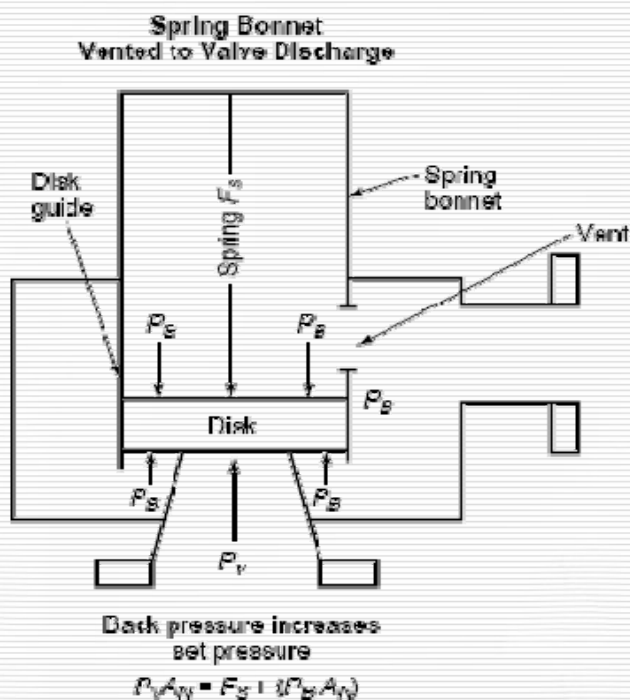


Figure 22—Typical Relationship Between Lift of Disk in a Pressure Relief Valve and Vessel Pressure

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$$A_D > A_N$$

A_D = disk area,

A_N = nozzle seat area,

F_s = spring force,

P_v = vessel pressure in pounds per square inch gauge,

P_B = superimposed back pressure, in pounds per square inch gauge.

Figure 22—Typical Effects of Superimposed Back Pressure on the Opening Pressure of Conventional Pressure Relief Valves

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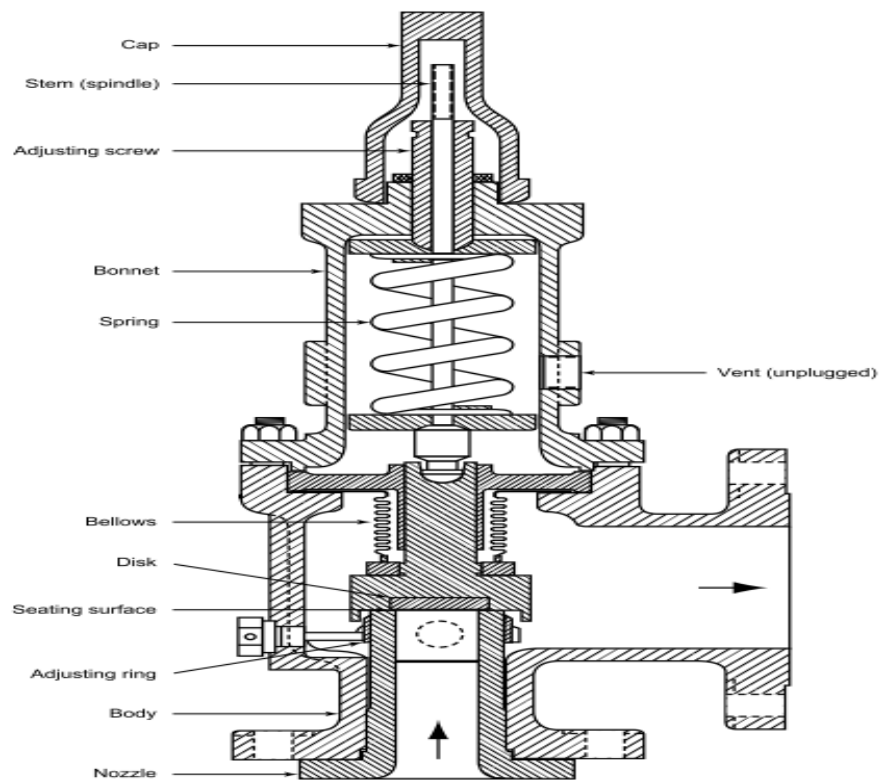
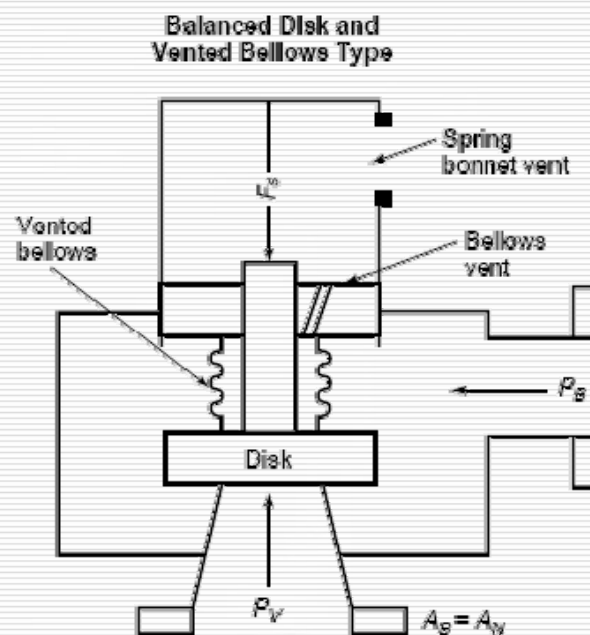


Figure 3—Balanced-Bellows Pressure Relief Valve

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Balanced valves can typically be applied where the total back pressure (superimposed plus built-up) does not exceed approximately 50% of the set pressure _ API 520

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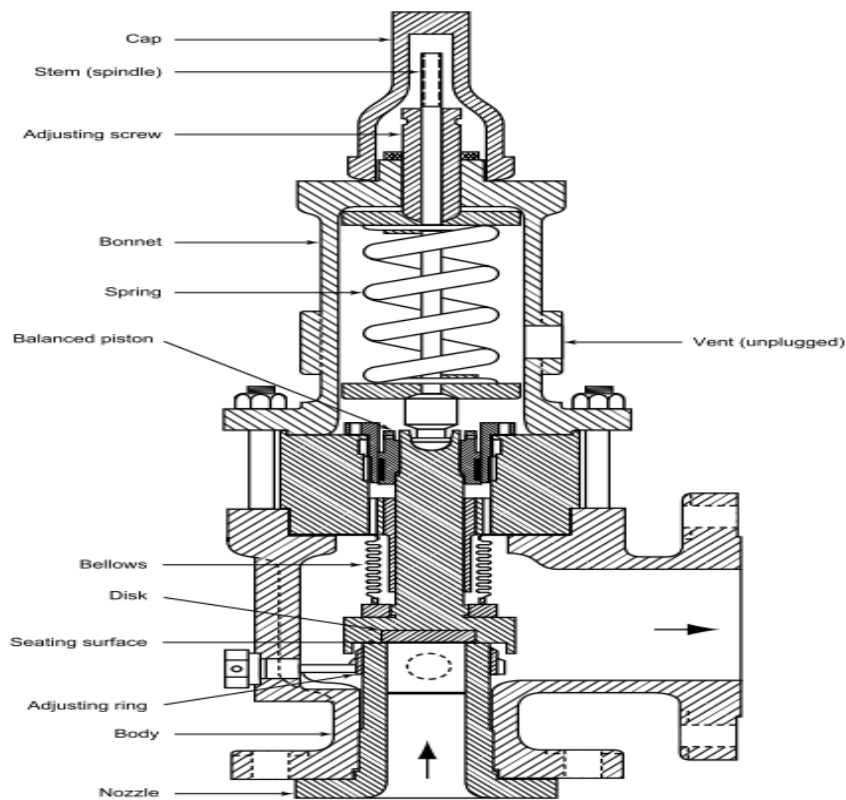
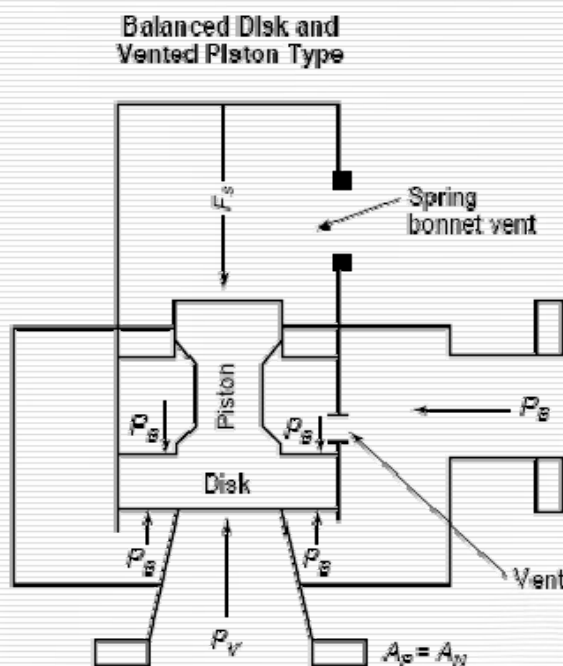


Figure 4—Balanced-Bellows Pressure Relief Valve with an Auxiliary Balanced Piston

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A_s = effective bellows area,

A_D = disk area,

A_N = nozzle seat area,

A_P = piston area (top),

F_s = spring force,

P_V = vessel gauge pressure,

P_g = superimposed back pressure, in pounds per square inch gauge,

P_S = set pressure, in pounds per square inch gauge.

Note: In this figure, $P_V = P_S$; $(P_V)(A_N) = F_s$ (typical); and $P_S = F_s/A_N$.

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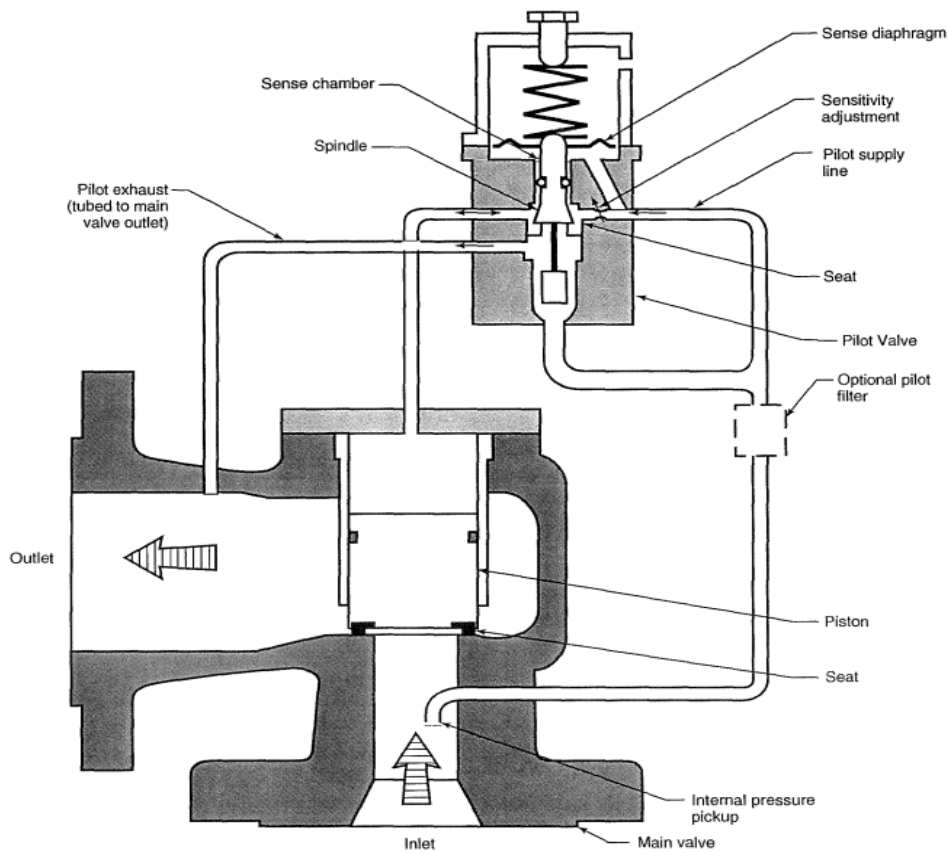


Figure 12—Modulating Pilot-operated Valve (Flowing-type)

流體中含水
或有固體者
，不適用。

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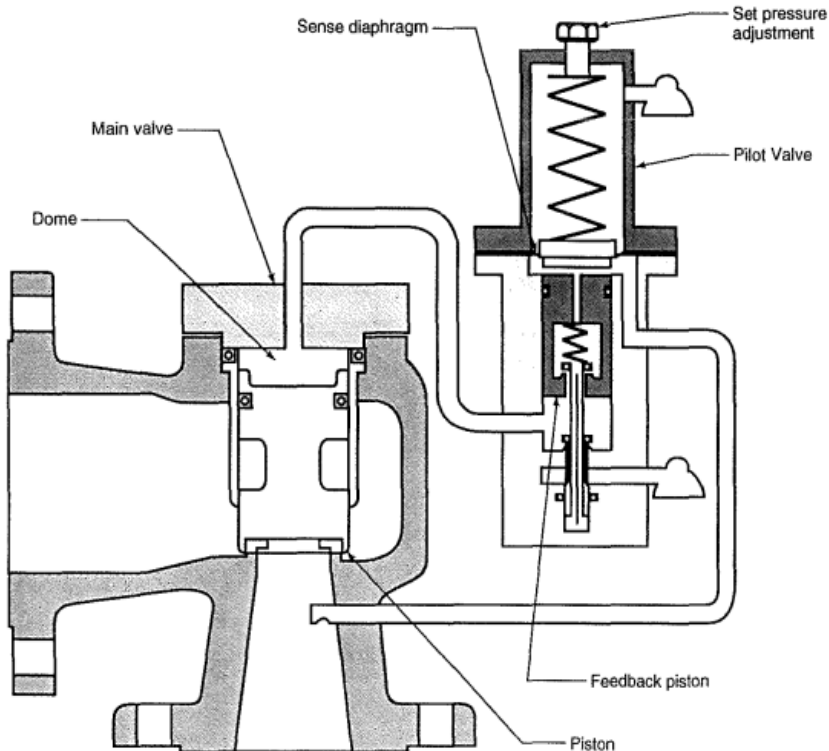


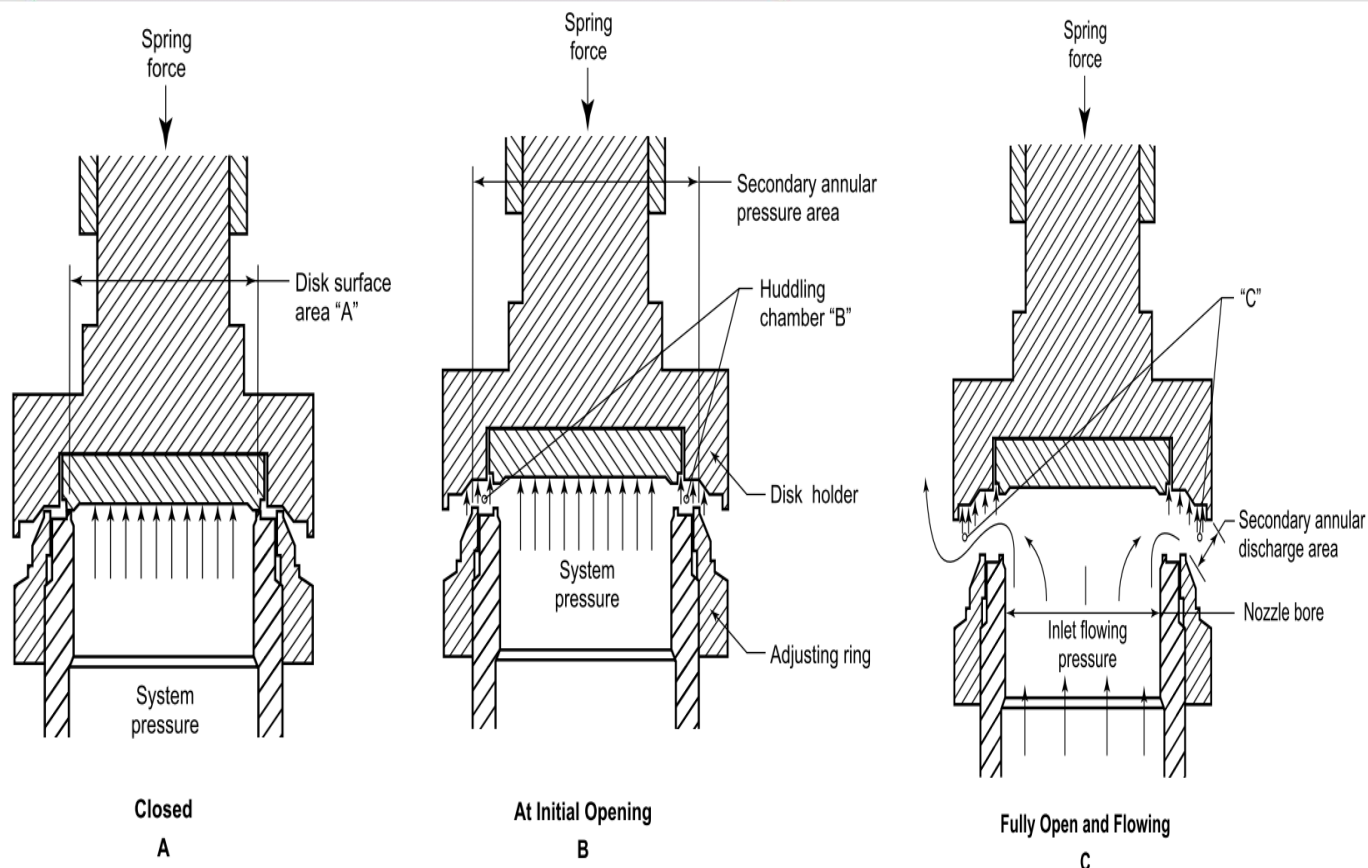
Figure 13—Pilot-operated Relief Valve with a Non-flowing Modulating Pilot Valve

適用於液體
、氣體、或
兩相流。

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Rupture Disks - 用途

1. 可以作為non-reclosing 之過壓保護設施。
2. 位於PRV之上游：
 - 1) 可免除VOC排放，PRV腐蝕之保護，可減少PRV之維修。
 - 2) Burst Pressure與PRV Set Pressure要一致，最好是組合在一起。
 - 3) 與PRV之間要有Free Vent與Pressure Indicator，可以自動告知Rupture Disk是否已經破裂。
 - 4) 製程流體容易聚合之情況，保護PRV被黏住。
3. 位於PRV之下游：

可以保護PRV免受下游流體之影響。
4. 高腐蝕環境之應用：

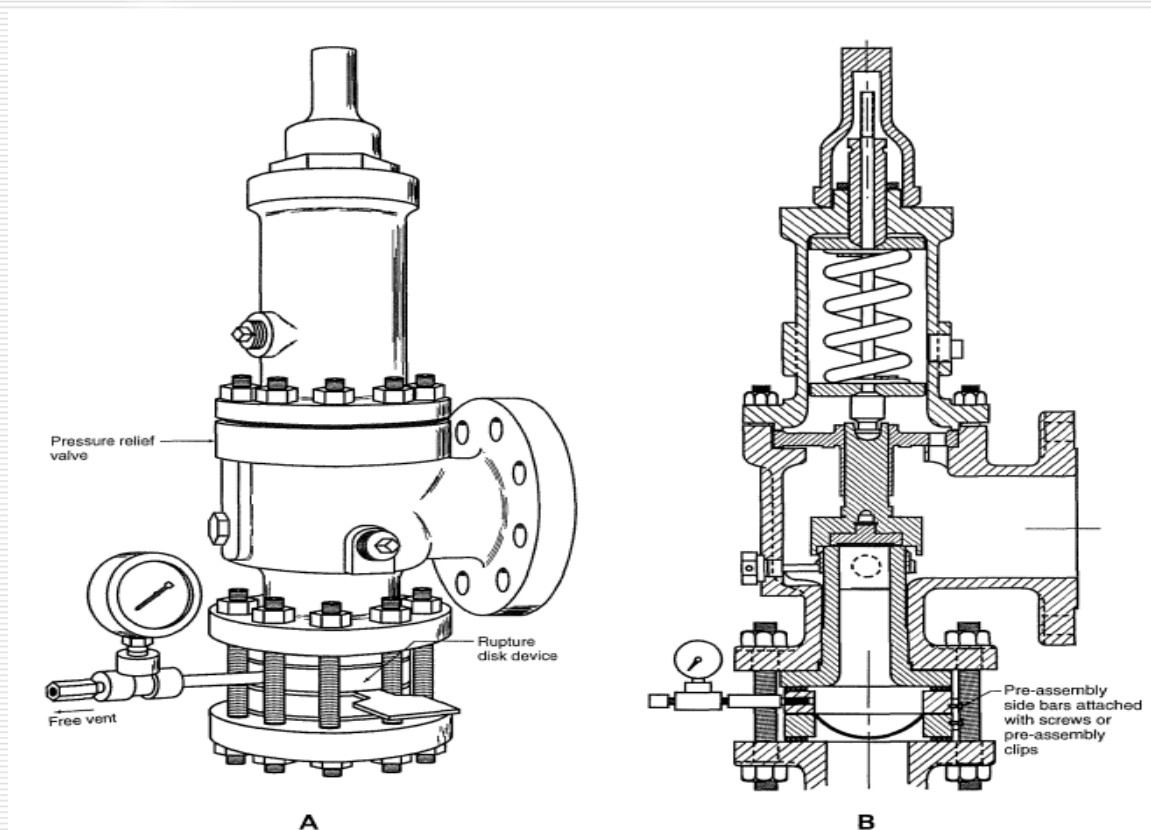
通常設置兩片，中間要有Free Vent與Pressure Indicator。

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1. 真空系統或有背壓存在時，需搭配Support以避免Reverse Flexing。
2. Marked burst pressure 有可能落在製造商之design range之內，其製造上限不可超出MAWP。
3. Rupture Disk之選用比較複雜，建議要多多與具有信譽之製造商溝通，深入了解產品之特性 (design range & operating ratio)，以購買到最適當之產品。
4. 有背壓存在時要特別留意，高估背壓時會導致Rupture Disk提早爆開。(請參考補充資料, Fig. 28)



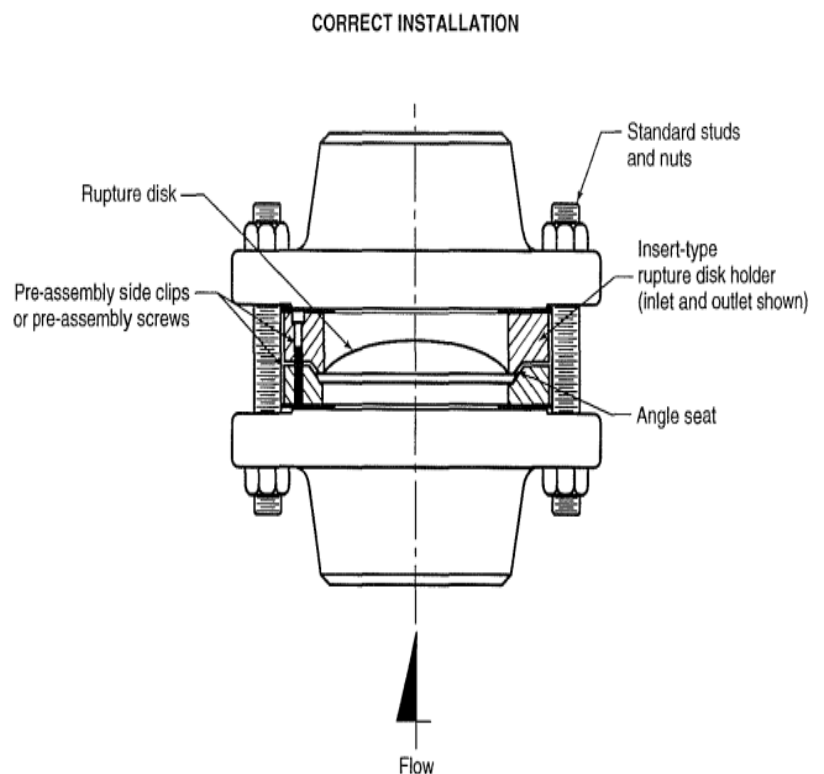
REVERSE BUCKLING BURSTING DISCS		Operating Pressure Ratio @ 20°C (ii)
	MN/MO	0.95
	RBH/RBF	0.95
	LRBH	0.95
	SRBH	0.90
	RBX	0.90
	GRB	0.90

FORWARD ACTING BURSTING DISCS		Operating Pressure Ratio @ 20°C (ii)
	NT/NR	0.75
	NTG/NRG	0.80
	CS	0.80
	GR	0.80
	MONO BLOC	0.80

(ii) The **Operating or Working Ratio** is the ratio of the Working Pressure to the minimum tolerance Burst Pressure. Reverse Buckling discs can offer a higher capability than Conventional Tensile-loaded discs. This ratio can be affected by disc material and operating temperature.

Solid Metal
(70% of Operating Ratio)

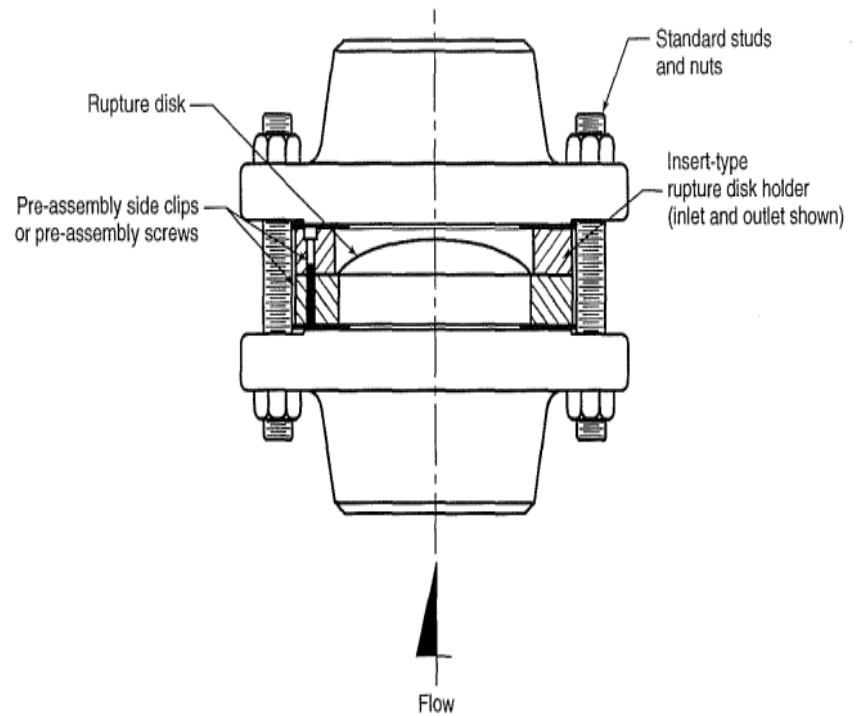
(Tension Loaded)



Score
(85 - 90% Operating Ratio)



(Tension Loaded)



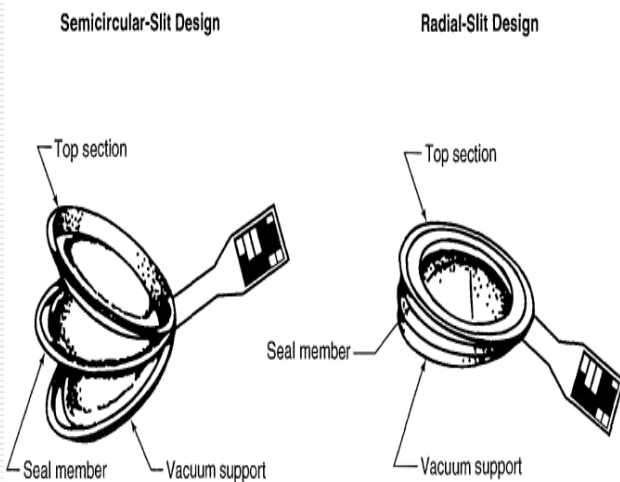
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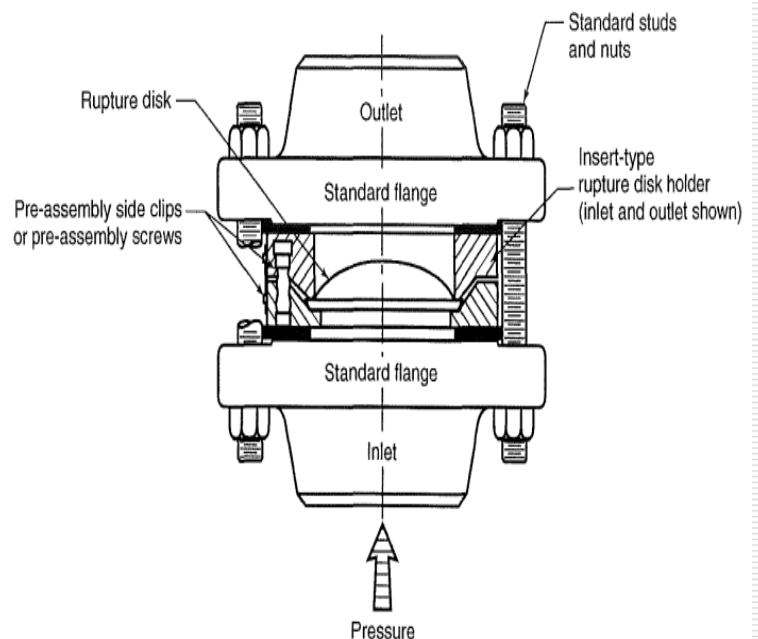
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Composite
(80% Operating Ratio)

(Tension Loaded)



CORRECT INSTALLATION



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Rupture Disks - Reverse-Acting

(Compression Loaded)

Knife Blades

(90- 95% Operating Ratio)

- 不會破裂成碎片，故適合安裝在於PRV之上游。

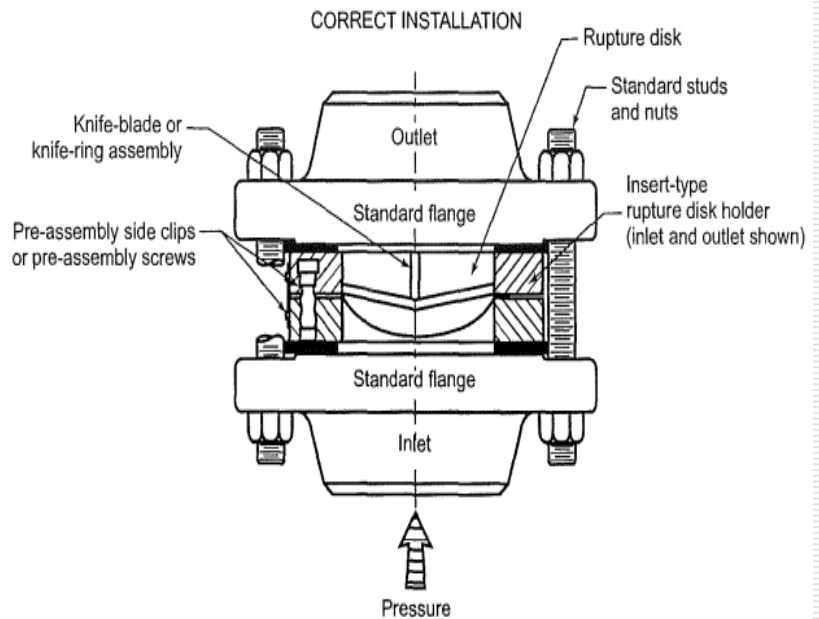


Figure 23—Reverse-acting Rupture Disk with Knife Blades

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Rupture Disks - Reverse-Acting

(Compression Loaded)

Knife Blades

(90- 95% Operating Ratio)

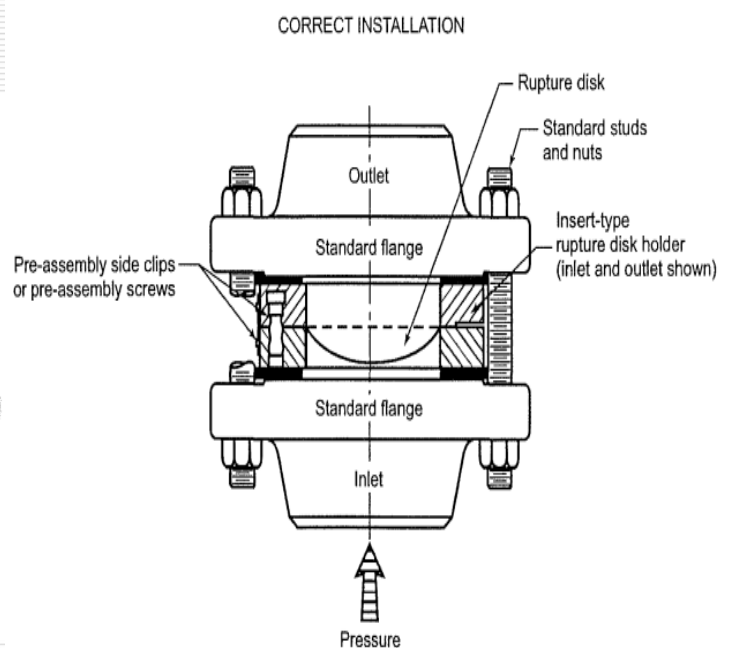


Figure 24—Reverse-acting Scored Rupture Disk

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(Shear Loaded)

Graphite
(80% Operating Ratio)

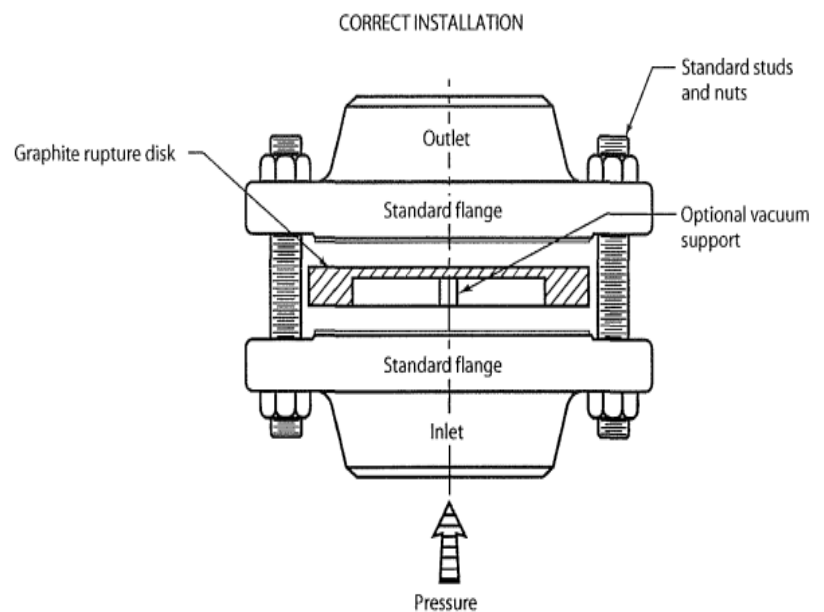
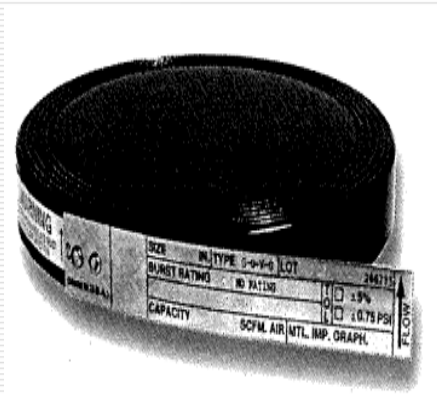


Figure 25—Graphite Rupture Disk

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安全排放量計算 — 基本觀念

- ◆ 對於各種可能造成過壓的情況，只需考慮一個獨立的因素，不需同時考慮兩個或兩個以上不相關的原因。

(此為API 521設計之基本觀念)

- ◆ 過壓大小的決定原則：

- a) 通常不一定是單一設備，設備之間相互連通，彼此間無阻閥或其他裝置阻隔者要一起考慮。
- b) 先就各種可能造成過壓之原因計算其排放量，取其最大者做為此設備或系統的安全閥決定量。

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- 1.) External fire
- 2.) Blocked outlets
- 3.) Utility failures (Loss of cooling water)
- 4.) Loss of cooling duty (top, intermediate reflux)
- 5.) Thermal Expansion
- 6.) Abnormal heat input
- 7.) Abnormal vapor input
- 8.) Loss of absorbent flow
- 9.) Entrance of highly volatile material
- 10.) Accumulation of noncondeserables
- 11.) Valve malfunction (Control Valve fail open or closed)
- 12.) Process control failure
- 13.) Equipment mechanical failure
- 14.) Exchanger tube rupture
- 15.) Upstream relieving
- 16.) Runaway chemical reaction
- 17.) Human error (such as inadvertent valve operation)

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Condensate- induced Hammer :

1. 通常發生在冷凝水系統， **isolated steam bubble**遇到冷的**condensate**後**bubble**爆開，壓力瞬間升高，嚴重者會造成管線之破裂，因為是瞬間產生，故此一形式之過壓現象無法用**PSV**來保護。
2. 需適當地規劃配管系統，可以避免此一現象，原則上
 - 不同壓力等級冷凝水，避免共用同一主管線
 - 管線要Free Drain不能有Pocket發生。

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真空疏解 (Vacuum Relief) :

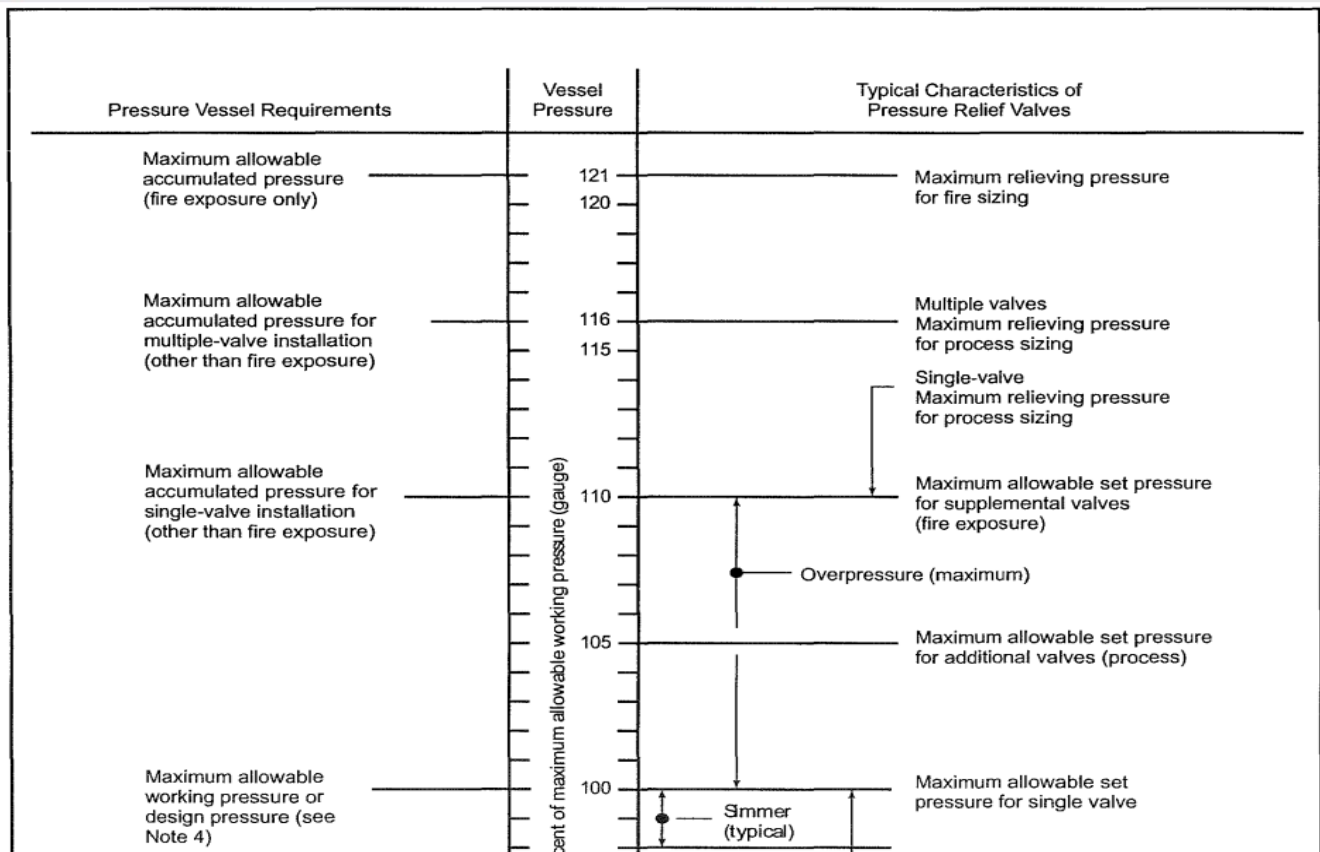
原因

- 1) 當設備的流出量大於流入量就會有負壓產生。
- 2) 當溫度降低導致設備內氣體收縮，也會造成真空。
- 3) 其他吸附與化學反應導致設備內氣體減少。

對策

- 1) 操作程序之規劃。
- 2) 提高設備之設計壓力 (耐負壓)。
- 3) 增設呼吸閥保護或壓力平衡管。

- 對於設備僅裝設單獨一安全閥，其設定壓力應不超過MAWP。
- 若裝設兩個PRV，其第二個的設定壓力應設定在不超過MAWP的105%，為防止因火災或其他無法預知的外來熱源而裝設的Supplemental-Device，其設定壓力應設定在不超過MAWP的110%。
- 安全閥在壓力達Accumulation時 - 閥全開，而在計算安全閥的排放量時，必須是以在其排放條件下(at relieving condition)，也就是說，若排放原因是火災，其排放條件就是在壓力等於121% SP，若為其他製程原因引起的釋壓排放，且裝設單一安全閥時，其排放條件就是在壓力等於110% SP，若裝兩個安全閥，則為116% SP。
- 預期可能的最高操作壓力和MAWP間通常有10%的安全係數，但此數值可較高或較低。
- 安全閥在約98% SP開始打開，在約92.5% SP時全關。



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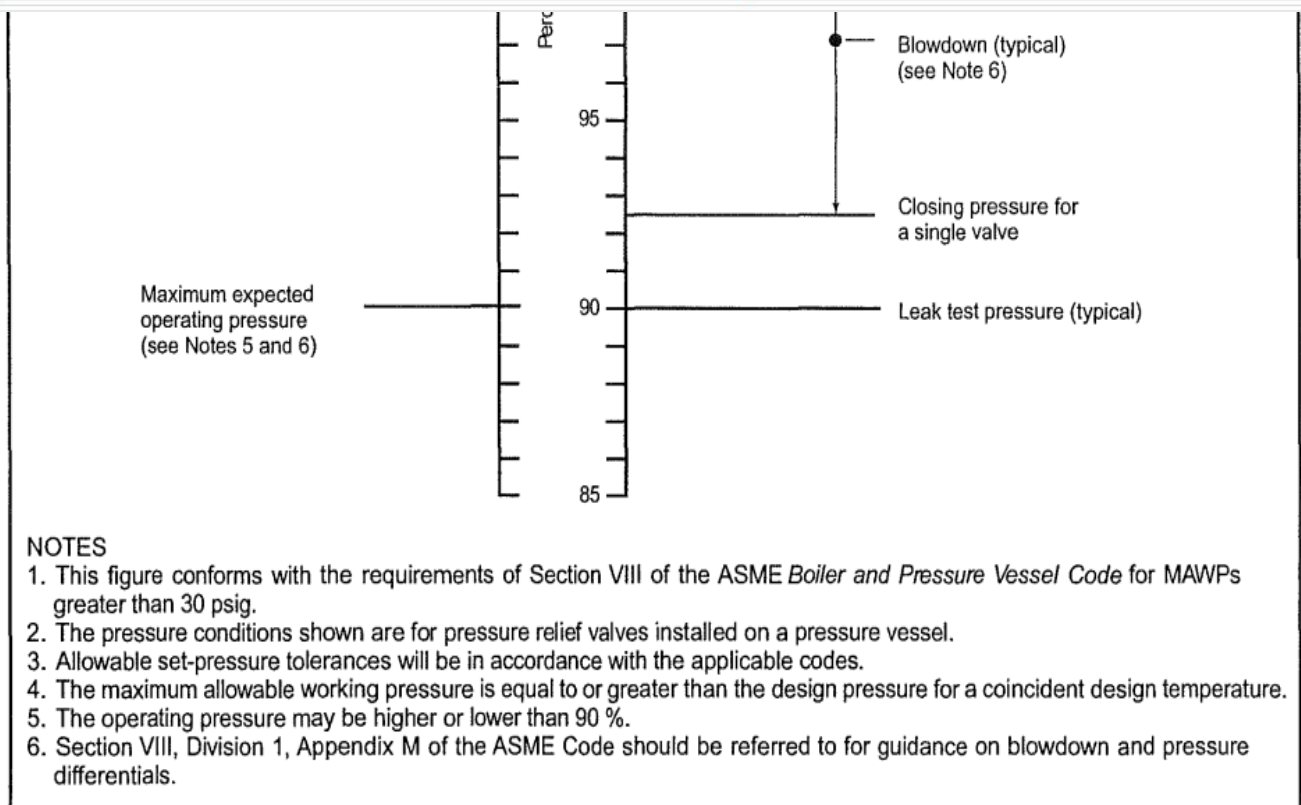
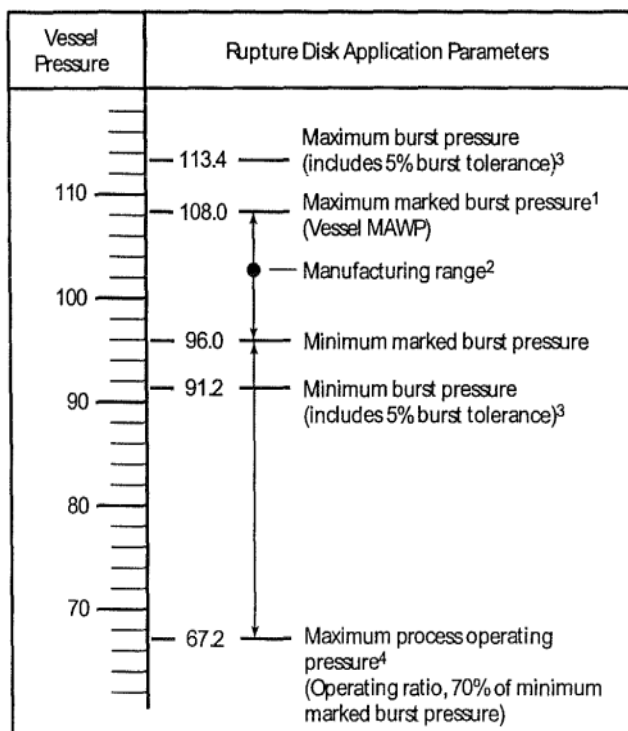


Figure 15—Pressure Level Relationships for PRVs

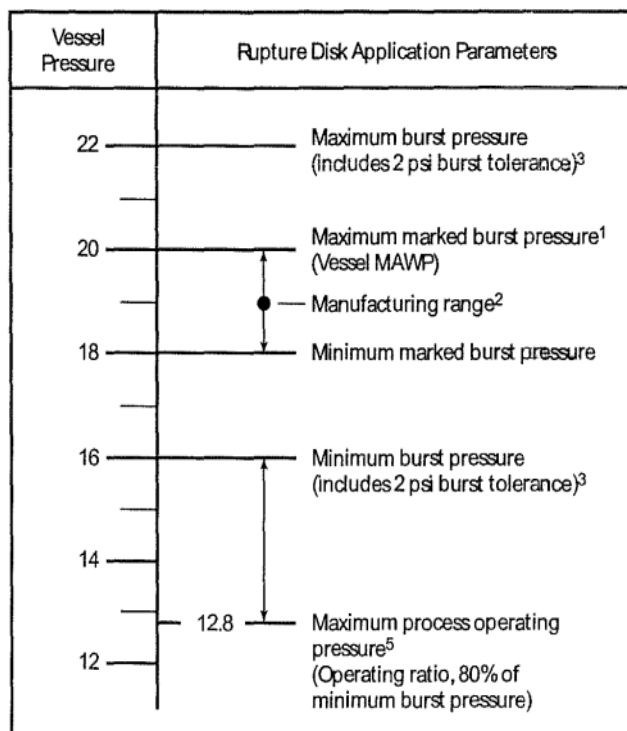
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A. Example of a rupture disk with a specified burst pressure of 100 psig, manufacturing range of +8/-4%, burst tolerance of $\pm 5\%$, and a 70% operating ratio.

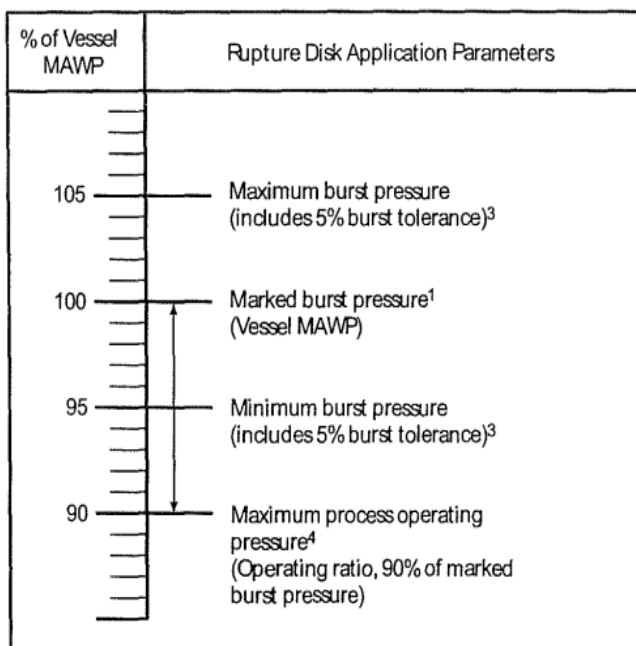


C. Example of a rupture disk with a specified burst pressure of 20 psig, manufacturing range of +0/-10%, burst tolerance of ± 2 psig, and an 80% operating ratio.

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B. Example of a rupture disk with a specified burst pressure of 100 psig, zero manufacturing range, burst tolerance of $\pm 5\%$, and a 90% operating ratio.

NOTE

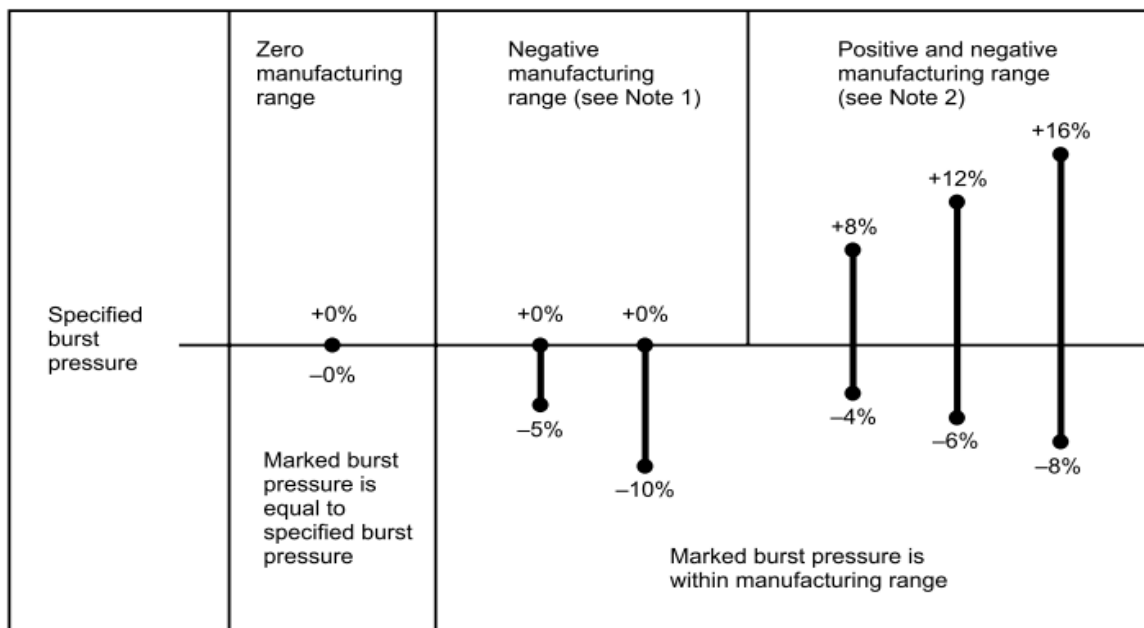
1. See Figure 18 for limits on marked burst pressure.
2. Marked burst pressure may be any pressure within the manufacturing range, see Figure 27.
3. For marked burst pressures above 40 psig, the burst tolerance is $\pm 5\%$. For marked burst pressures at 40 psig and below, the burst tolerance is ± 2 psi.
4. For marked burst pressures above 40 psig, the maximum process operating pressure is calculated by multiplying the minimum marked burst pressure by the operating ratio.
5. For marked burst pressures at 40 psig and below, the maximum process operating pressure is calculated by subtracting the burst tolerance from the minimum marked burst pressure, then multiplying the difference by the operating ratio.

Figure 26—Rupture Disk Application Parameters Assuming No Superimposed Backpressure

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Notes:

1. The marked burst pressure will not exceed the specified burst pressure.
2. Positive manufacturing range may result in a marked burst pressure exceeding the specified burst pressure.

Figure 27—Common Types of Manufacturing Ranges and Corresponding Burst Pressure Marking

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Table 1—Set Pressure and Accumulation Limits for Pressure Relief Devices

Contingency	Single Device Installations		Multiple Device Installations	
	Maximum Set Pressure %	Maximum Accumulated Pressure %	Maximum Set Pressure %	Maximum Accumulated Pressure %
Non-fire Case				
First relief device	100	110	100	116
Additional device(s)	—	—	105	116
Fire Case				
First relief device	100	121	100	121
Additional device(s)	—	—	105	121
Supplemental device	—	—	110	121
NOTE All values are percentages of the maximum allowable working pressure.				

5.4.2.1 Single-device Installation

5.4.2.1.1 In accordance with the requirements of the ASME Code, accumulated pressure shall be limited to 110 % of the maximum allowable working pressure, MAWP, in vessels that are protected by a single pressure relief device sized for operating (non-fire) contingencies. The set pressure of the device shall not exceed the MAWP.

5.4.2.1.2 The allowable accumulation is 3 psi (21 kPa) when the MAWP is between 15 psig and 30 psig (103 kPag and 207 kPag) in accordance with the ASME Code.

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Table 2—Example Determination of Relieving Pressure for Operating Contingencies for a Single Relief Device Installation

Characteristic	Value
Relief Device Set Pressure Equal to MAWP	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	110.0 (758)
Relief device set pressure, psig (kPag)	100.0 (689)
Allowable overpressure, psi (kPa)	10.0 (69)
Barometric pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	124.7 (860)
Relief Device Set Pressure Less Than MAWP	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	110.0 (758)
Relief device set pressure, psig (kPag)	90.0 (621)
Allowable overpressure, psi (kPa)	20.0 (138)
Barometric pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	124.7 (860)
NOTE The above examples assume a barometric pressure of 14.7 psia (101.3 kPa). The barometric pressure corresponding to site elevation should be used.	

5.4.2.2.1 A multiple-device installation requires the combined capacity of two or more pressure relief devices to alleviate a given overpressure contingency.

5.4.2.2.2 In accordance with the requirements of the ASME Code accumulated pressure shall be limited to 116 % of the maximum allowable working pressure, MAWP in vessels that are protected by multiple pressure relief devices sized for operating (non-fire) contingencies. The set pressure of the first device shall not exceed the MAWP. The set pressure of the additional device or devices shall not exceed 105 % of the MAWP.

5.4.2.2.3 The allowable accumulation is 4 psi (28 kPa) when the MAWP is between 15 psig and 30 psig (103 kPag and 207 kPag).

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Table 3—Example Determination of Relieving Pressure for Operating Contingencies for a Multiple Relief Device Installation

Characteristic	Value
First Relief Device (Set Pressure Equal to MAWP)	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	116.0 (800)
Relief device set pressure, psig (kPag)	100.0 (689)
Allowable overpressure, psi (kPa)	16.0 (110)
Barometric pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	130.7 (901)
Additional Relief Device (Set Pressure Equal to 105 % of MAWP)	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	116.0 (800)
Relief device set pressure, psig (kPag)	105.0 (724)
Allowable overpressure, psi (kPa)	11.0 (76)
Barometric pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	130.7 (901)
NOTE The above examples assume a barometric pressure of 14.7 psia (101.3 kPa). The barometric pressure corresponding to site elevation should be used.	

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Table 4—Example Determination of Relieving Pressure for Fire Contingencies for a Single Relief Device Installation

Characteristic	Value
Relief Device Set Pressure Equal to MAWP	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	121.0 (834)
Relief device set pressure, psig (kPag)	100.0 (689)
Allowable overpressure, psi (kPa)	21.0 (145)
Barometric pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	135.7 (936)
Relief Device Set Pressure Less Than MAWP	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	121.0 (834)
Relief device set pressure, psig (kPag)	90.0 (621)
Allowable overpressure, psi (kPa)	31.0 (214)
Barometric pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	135.7 (936)
NOTE The above examples assume a barometric pressure of 14.7 psia (101.3 kPa). The barometric pressure corresponding to site elevation should be used.	

Table 5—Example Determination of Relieving Pressure for Fire Contingencies for a Multiple Relief Device Installation

Characteristic	Value
First Relief Device (Set Pressure Equal to MAWP)	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	121.0 (834)
Relief device set pressure, psig (kPag)	100.0 (689)
Allowable overpressure, psi (kPa)	21.0 (145)
Barometric pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	135.7 (936)
Additional Relief Device (Set Pressure Equal to 105 % of MAWP)	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	121.0 (834)
Relief device set pressure, psig (kPag)	105.0 (724)
Allowable overpressure, psi (kPa)	16.0 (110)
Barometric pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	135.7 (936)
NOTE The above examples assume a barometric pressure of 14.7 psia (101.3 kPa). The barometric pressure corresponding to site elevation should be used.	

Table 6—Example Determination of Relieving Pressure for Fire Contingencies for a Supplemental Valve Installation

Characteristic	Value
First Relief Device (Set Pressure Equal to MAWP)	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	121.0 (834)
Relief Device set pressure, psig (kPag)	100.0 (689)
Allowable overpressure, psi (kPa)	21.0 (145)
Barometric Pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	135.7 (936)
Supplemental Relief Device (Set Pressure Equal to 110 % MAWP)	
Protected vessel MAWP, psig (kPag)	100.0 (689)
Maximum accumulated pressure, psig (kPag)	121.0 (834)
Relief Device set pressure, psig (kPag)	110.0 (758)
Allowable overpressure, psi (kPa)	11.0 (76)
Barometric Pressure, psia (kPa)	14.7 (101)
Relieving pressure, P_1 , psia (kPa)	135.7 (936)
NOTE The above examples assume a barometric pressure of 14.7 psia (101.3 kPa). The barometric pressure corresponding to site elevation should be used.	

在計算安全閥的排放量及相關物性時，必須是基於其排放條件下，就是指其壓力達排放壓力(= Set Pressure + Overpressure)的條件。

<u>Contingency</u>	<u>Relieving Pressure</u>
Non-fire single valve	110% Set pressure
Non-fire Multiple valve	116% Set pressure
Fire Case	121% Set pressure

Table 2 — Guidance for required relieving rates under selected conditions

Item No.	Condition	Liquid-relief guidance ^a	Vapour-relief guidance ^a
1	Closed outlets on vessels	Maximum liquid pump-in rate	Total incoming steam and vapour plus that generated therein at relieving conditions
2	Cooling-water failure to condenser	—	Total vapour to condenser at relieving conditions
3	Top-tower reflux failure	—	Total incoming steam and vapour plus that generated therein at relieving conditions less vapour condensed by sidestream reflux
4	Sidestream reflux failure	—	Difference between vapour entering and leaving equipment at relieving conditions
5	Lean-oil failure to absorber	—	None, normally
6	Accumulation of non-condensables	—	Same effect in towers as found for Item 2; in other vessels, same effect as found for Item 1

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7	Entrance of highly volatile material	—	—
	Water into hot oil	—	For towers, usually not predictable
	Light hydrocarbons into hot oil	—	For heat exchangers, assume an area twice the internal cross-sectional area of one tube to provide for the vapour generated by the entrance of the volatile fluid due to tube rupture
8	Overfilling storage or surge vessel	Maximum liquid pump-in rate	—
9	Failure of automatic controls	—	Analyse on a case-by-case basis
10	Abnormal heat or vapour input	—	Estimated maximum vapour generation including non-condensables from overheating
11	Split exchanger tube	Liquid entering from twice the cross-sectional area of one tube	Steam or vapour entering from twice the cross-sectional area of one tube; also same effects found in Item 7 for exchangers
12	Internal explosions	—	Not controlled by conventional relief devices but by avoidance of circumstances
13	Chemical reaction	—	Estimated vapour generation from both normal and uncontrolled conditions; consider two-phase effects

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14	Hydraulic expansion:		
	Cold-fluid shut in	See 5.14	—
	Lines outside process area shut in	See 5.14	—
15	Exterior fire ^b	See 5.15.3.3	Estimated by the methods given in 5.15.2.2 or 5.15.3.2
16	Power failure (steam, electric, or other)	—	Study the installation to determine the effect of power failure; size the relief valve for the worst condition that can occur
	Fractionators	—	Loss of all pumps, with the result that reflux and cooling water would fail
	Reactors	—	Consider failure of agitation or stirring, quench or retarding stream; size the valves for vapour generation from a runaway reaction
	Air-cooled exchangers	—	Fan failure; size valves for the difference between normal and emergency duty
	Surge vessels	—	Maximum liquid inlet rate
^a Consideration can be given to the reduction of the relief rate as the result of the relieving pressure being above operating pressure. ^b Guidance on fire relief is given in Annex A.			

A) Fire Case排放量計算依據：

- 設計壓力 > 15 psig (1.034 barg)，依照API 521的內容。
- 設計壓力 ≤ 15 psig (1.034 barg)，依照API 2000的內容。
- 新版的API 520已經將如何計算Fire Case排放量的相關內容完全刪除。
- NFPA 30中的相關內容在Annex B，基本上內容摘錄自API 521，但沒有API 521的詳細。

- ◆ 只需考慮火源上方7.6m (25ft)以下的設備受熱面積，
若樓板為水泥或花紋鐵板(無縫隙)則高度需要重新歸零。
- ◆ 排放溫度經常超過設備設計溫度，若升高的溫度可能造成
設備損壞必須另外考慮增加保護措施
(詳細內容請參閱 API 521 5.15.4)
- ◆ Fire zone : 230~460 m² (API 521)

吸熱量 (Q) 與排放量 (W) 的方程式：

設備內盛裝液體時：

$$Q = C_f * F * A^{0.82} \quad (\text{設備內盛裝液體時所吸收之熱功率，kcal/hr})$$

$$W = Q / \lambda \quad (\lambda \text{ 疏解壓力下液體之Latent Heat，kcal/kg})$$

C_f 為常數=37160 or 61015，F為環境因子

設備內盛裝氣體時：

$$Q = F_G * C_f * F * A^{0.82} \quad (\text{設備內充滿氣體時，} F_G \text{為修正係數=0.3})$$

$$W = Q / (C_p * \Delta T)$$

(λ 疏解壓力下氣體之 C_p 為定壓比熱，kcal/kg-°C； ΔT 為 上升之溫度°C)

對於設計壓力 15 psig 以上的設備，於火災情況下，設備吸熱量的方程式依該設備是否有適當的排放途徑及滅火設備可分為二：

Q (Btu/hr), A(ft ²)	Q (W), A(m ²)	Q (kcal/hr), A(m ²)
火災發生後，有即時的救火行動及設備內易燃物質的排放 (there are prompt firefighting efforts and drainage of flammable materials away from the vessels)		
$Q = 21,000 F A^{0.82}$	$Q = 43,200 F A^{0.82}$	$Q = 37,160 F A^{0.82}$
無適當的排放途徑及滅火設備		
$Q = 34,500 F A^{0.82}$	$Q = 70,900 F A^{0.82}$	$Q = 61,015 F A^{0.82}$
Q = total heat absorption (input) to the wetted surface F = environmental factor A = total wetted area		

“化工製程安全設計”一書中有關設備內液體接觸的表面積的估算如下，但最好是根據實際設備高度與液位來計算。

- 直立筒形桶、槽或容器：底部表面積與85%直立部分外殼表面積之和。
- 橫臥筒形桶、槽或容器：75%外殼表面積。
- 球形桶、槽或容器：80%外殼表面積。

表面積計算公式：

設備種類	高度	垂直安裝	水平安裝
Liquid-full	All<7620mm	$\pi D L + 1.084 \times D^2 \times 2$	
Vessel	TTL>7620mm BTL<7620mm	$\pi D (L - 7620) + 1.084 \times D^2$	$(1 - \cos^{-1}((7620 - BTL)/D/0.5 - 1) \div \pi) \times (\pi D L + 1.084 \times D^2 \times 2)$
	BTL>7620mm Btm<7620mm	$1.084 \times D^2$	—
Surge Drum, etc.	NLL<7620mm	$\pi D L \times 85\% + 1.084 \times D^2$	$(\pi D L + 1.084 \times D^2 \times 2) \times 75\%$
	NLL>7620mm BTL<7620mm	$\pi D (L - 7620) + 1.084 \times D^2$	$(1 - \cos^{-1}((7620 - BTL)/D/0.5 - 1) \div \pi) \times (\pi D L + 1.084 \times D^2 \times 2)$
	BTL>7620mm Btm<7620mm	$1.084 \times D^2$	—

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Tower	Btm tray <7620mm	$\pi \times D \times NLL + 1.084 \times D^2 + \pi \times D \times N \times \text{Weir height (for tray tower)}$
	NLL>7620mm BTL<7620mm	$\pi D (L - 7620) + 1.084 \times D^2$
	BTL>7620mm Btm<7620mm	$1.084 \times D^2$

Notes:

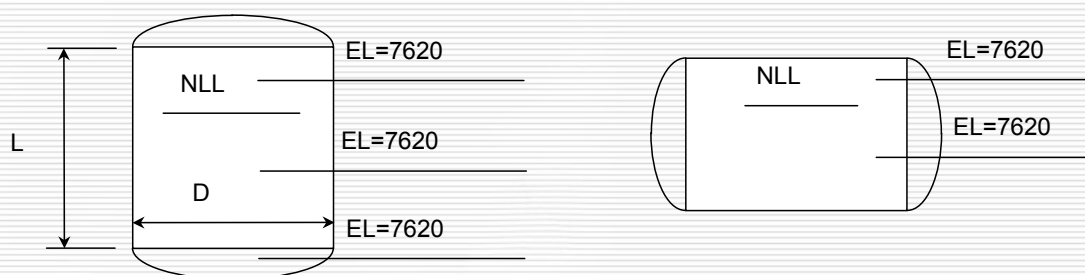
1. NLL : elevation of normal liquid level
BTL : elevation of bottom tangent line
TTL : elevation of top tangent line
N : 比 7620mm 低的塔板數
2. D: 設備內徑; L: 設備長度(TL-TL) (mm)
3. 2:1 橢圓端板面積為 $1.084 \times D^2(\text{mm}^2)$
4. 熱交換器視同 liquid-full vessel, 但必須就 PSV 所保護的是殼側或管側分別計算其接液面積

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5. 對於非滿液操作的設備之 NLL，在上述的計算中採較保守的原則，對直立設備 $NLL=85\% L$ ，對水平設備 $NLL=85\% D$ (表面積約為全部表面積的 74.68%)
6. 相對於 $EL=7620mm$ 的各種不同設備安裝高度，其接液面積的計算，為求明瞭請參考下圖



Environmental Factor (F)的訂定：

- 環境因子 (F) 與是否具有保溫材料或灑水設施有關，一般原則：
 $F = 1.0$: 表面無保溫材料，亦無灑水設施
 $F = 0.3$: 表面有保溫材料或灑水設施
 $F = 0.15$: 兼具有保溫材料及灑水設施
- API 521 & API 2000規定較為詳細，兩者內容基本上是一致的，API 2000中的有保溫的厚度和F factor的對照，對F factor的訂定更方便。

Hydraulic Expansion (疏解速率)

Hydraulic expansion是液體因為溫度的提高，造成體積的增加，液體體積膨脹的原因很多，最常見的有下列三種：

1. 配管或設備之進出口被關閉之情形下，液體因為受熱而膨脹造成壓力升高，進而促使管線或設備破裂釀成災害，熱源可以是蒸汽加熱之Coil、Tracing或火災，甚至陽光輻射熱。
2. 熱交換器之冷卻水側通常需要設置Thermal Relief Valve。
3. 封閉管線受日照輻射熱容易引起液體膨脹，也要設置Thermal Relief Valve。

◆ 因為排放時為液體狀態，雖然Thermal Relief Valve通常只要3/4"×1"即足夠，但最好還是要計算，除了安全檢查需要提供計算書，也需要確保Thermal Relief Valve之能力是足夠的。

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熱交換器從另一側加熱：

$$W = \beta_{liq} * U * A * \Delta T / C_p$$

W = expansion rate (kg/hr)

β_{liq} = thermal expansion coefficient of liquid (1/°C)

U = overall heat transfer coefficient,

(Nature Convection時，一般取200 kcal/m²-h-°C)

A = area for heat transfer (m²)

ΔT = LMTD = $(T_2 - T_1) / \ln[(T_{ext} - T_1) / (T_{ext} - T_2)]$ (°C)

T_{ext} = cold fluid inlet temperature (°C)

T_1 = cold fluid inlet temperature (°C)

T_2 = relieving temperature at relieving pressure (°C)

C_p = heat capacity of liquid (kcal/kg-°C)

疏解速率之計算及 β_{liq} 值參考 附件A：Chemical Properties Handbook, Chapter 28)

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$$\beta_{liq} = a*(1-T/T_c)^m$$

a , m = regression coefficients for chemical compound

T = temperature (°K)

T_c = critical temperature (°K)

日光照射加熱：

$$W = \beta_{liq} Q / C_p$$

$$Q = 2610A(65-T) \lambda / \sigma$$

Q = heat input from solar radiation (kcal/hr)

λ = thermal conductivity of insulation (kcal/m-h-°C)

σ = thickness of insulation (mm)

- 氣、液體兩相流之疏解計算仍在發展中，在”化工製程安全設計”(張一岑 著)之第3 章有詳細之介紹。
- 下列情形列需考慮氣、液體兩相流：

1) 反應失控：(Chemical Reaction)

- ◆ API 521 建議參考DIERS (Design Institute for Emergency Relief System)之方法來計算。

5.13 Chemical reaction

5.13.1 The methodology for determining the appropriate size of an emergency vent system for chemical reactions has been established by DIERS (Design Institute for Emergency Relief Systems) [38], [39], [40], [41], [42].

The DIERS methodology is based on the following:

- a) defining the design-basis upset conditions for the reaction system;
- b) characterizing the systems through bench-scale tests simulating the design-basis upset conditions;
- c) using vent-sizing formula that account for two-phase gas/liquid vent flow.

2) 火災時：

- ◆ 疏解裝置低於設備內部液面高度時。
- ◆ 疏解裝置雖然高於設備設備內部液位 $> 85\%$ 或黏度 $> 1 \text{ cps}$ ，且會產生許多泡沫時。

3) 程序失控：

宜分析失控後之異常現象，才能做決定，通常 Liquid/Vapor 速率比 > 0.15 時，宜考慮兩相流疏解。

4) 盛裝過熱液體之設備：

(疏解壓力－液體飽和壓力) < 0.6 (疏解壓力－排放管下游壓力) 時，液體流經疏解裝置之 Orifice 時會氣化，產生兩相流。

- 氣、液體兩相流較氣體或液體需要更大之疏解面積而言，故採用傳統單相流計算公式所得到之 PRV，其排放能力將不足。

1. 文獻數據與熱化學計算（理論或軟體—CHETAH）。

2. 經由實驗取得資料：

1) DSC (Differential Scanning Calorimetry)

2) DTA (Differential Thermal Analysis)

3) IET (Insulated Exotherm Test)

4) DPT (Decomposition Pressure Test)

5) Dewar Calorimeter

6) Iso-peribolic Calorimeter

7) Power Compensation Calorimeter

8) Heat Flow Calorimeter

9) Vent Sizing Package (VSP)

— 能提供最接近實際工場反應失控與 Vent Sizing 之數據，為 DIERS 所採用。

NFPA 68 有詳細之規定與說明，可分為兩大類（以**0.1bar**作為分界）

Low Strength Enclosures :

- 1) Pred (pressure-reduced) Not to exceed 1.5 psi, or 0.1 bar.
- 2) Low Strength Enclosures cannot have a vent duct.
- 3) Pred (pressure-reduced) should exceed Pstat by 0.35-psi (0.02-bar).
- 4) When the fundamental burning velocity of the gas exceeds 1.3 times that of propane
- 5) alternate methods of protection should be used.
- 6) When the vent area is restricted to one end of an elongated enclosure the ratio of the
- 7) length to diameter should be limited to three. Use the effective diameter for cross-sections
- 8) other than circular or square.

High Strength Enclosures :

- 1) Pred (pressure-reduced) > 0.1 bar (1.5-psi)
- 2) Pred (pressure-reduced) ≤ 2 bar (29-psi) and at least 0.05 bar > Pstat
- 3) Kg ≤ 550 bar-m/sec
- 4) Pstat ≤ 0.5 bar (7.5-psi)
- 5) Enclosure Volume ≤ 1000 cubic meter (35314.7 cubic feet)

Low Strength Enclosures : (<0.1 bar)

NFPA建議採用(Swift & Epstein)公式：

$$A = \frac{\frac{A_s \lambda S_u \rho_u}{C_0 G} \left\{ \left(\frac{P_m}{P_0} \right)^{1/\gamma} - 1 \right\}}{\left(\frac{P_f}{f_0} - 1 \right)^{\frac{1}{2}}}$$

可以簡化為下式：

$$A = \frac{C_1 A_s}{\sqrt{P}}$$

A = 疏解面積需求 (m²)

A_s = 結構的內表面積 (m²)

C₀ = 排放係數

λ = 亂流增加因數

S_u = 層流 (Laminar Flow) 的燃燒速率 (m/s)

f_u = 未燃燒的氣體密度 (kg/m³)

G = 單位面積質量速率 (kg/m²·s)

P_M = 疏解前最高爆炸壓力 (kpa)

P₀ = 最初壓力 (kpa)

P_f = 疏解時最高壓力 (kpa)

γ = 定壓比熱對定容比熱之比

Low Strength Enclosures : (<0.1 bar)

Swift & Epstein公式可以簡化為下式：

$$A = \frac{C_1 A_s}{\sqrt{P}}$$

P = 結構內部所能承受的最大壓力 (kpa)

C₁ = 排放常數 ($\sqrt{\text{kpa}}$)

燃燒物質	C* ($\sqrt{\text{kpa}}$)
甲烷	0.37
氨氣 (不含水份)	0.13
直鏈烴類氣體 (甲烷除外) 或基本燃燒速率 (Fundamental Burning Velocity) 低於丙烷燃燒速率1.3倍的氣體	0.45
St-1塵粒	0.26
St-2塵粒	0.30
St-3塵粒	0.51

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High Strength Enclosures : (> 0.1 bar)

可以由德國工程師學會 (VDI) 公式求出：

$$A_v = a V^{\frac{2}{3}} (K_{st})^b (P_M)^{-c}$$

$$K_{st} = \left(\frac{dP}{dt} \right)_{\text{最大值}} V^{\frac{1}{3}}$$

$$a = 5.73 \times 10^{-4} \exp (2P_s)$$

$$b = 0.978 \exp (-1.048P_s)$$

$$c = 0.687 \exp (0.226P_s)$$

A_v = 疏解面積需求 (m²)

V = 內部空間體積 (m³)

K_{st} = 塵粒突燃 (爆炸) 常數 (巴·公尺/秒, bar·m/s)

P_M = 疏解前最高爆炸壓力 (巴, bar)

P_s = 疏解裝置開啓壓力 (巴, bar)

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High Strength Enclosures : (> 0.1 bar)

疏解面積需求也可以下圖得到：

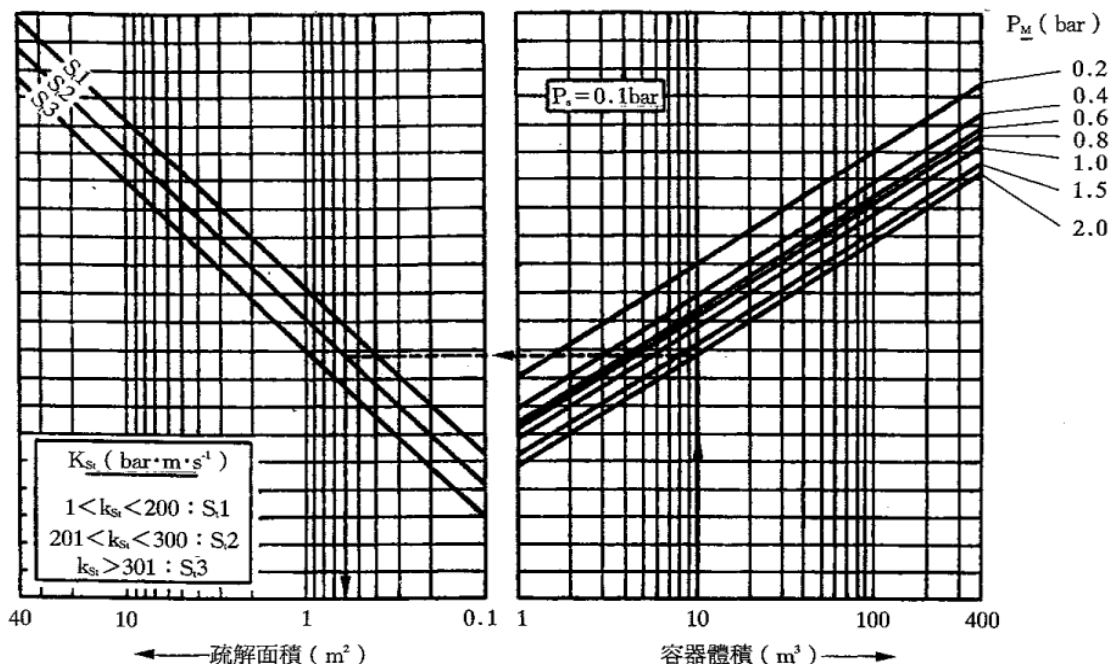


圖2-13 設定壓為0.1巴 (bar) 時，疏解面積、容器體積與塵粒爆炸常數的相互關係圖 [3]

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High Strength Enclosures : (≥ 0.2 bar)

疏解面積需求也可以下圖得到：

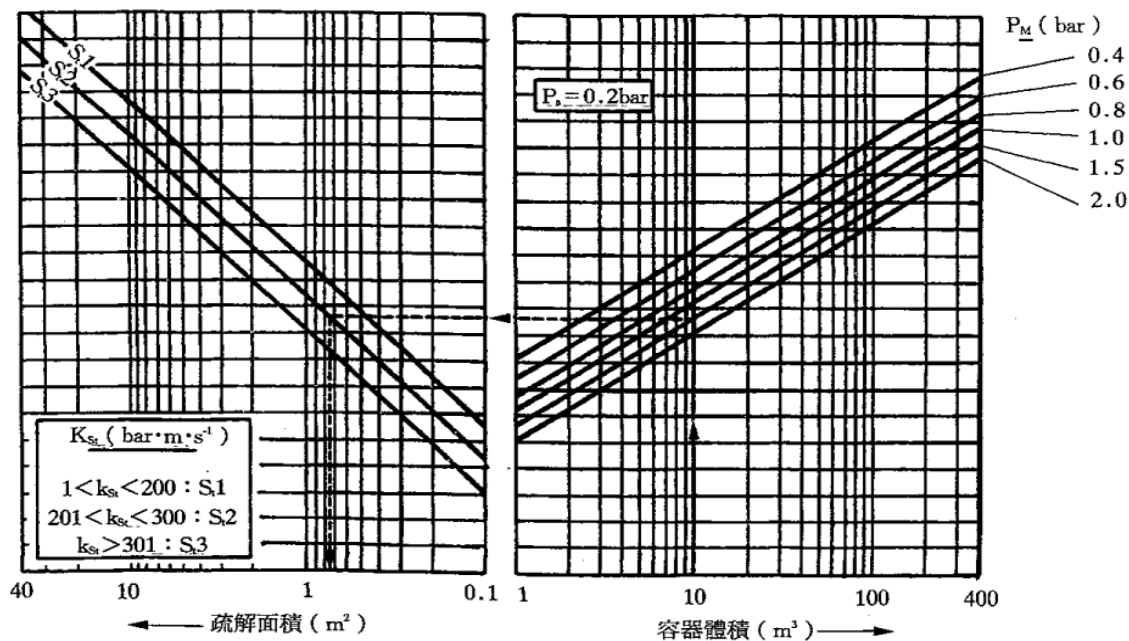


圖2-14 設定壓為0.2巴 (bar) 時，疏解面積、容器體積與塵粒爆炸常數的相互關係圖 [3]

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High Strength Enclosures : (≥ 0.5 bar)

疏解面積需求也可以下圖得到：

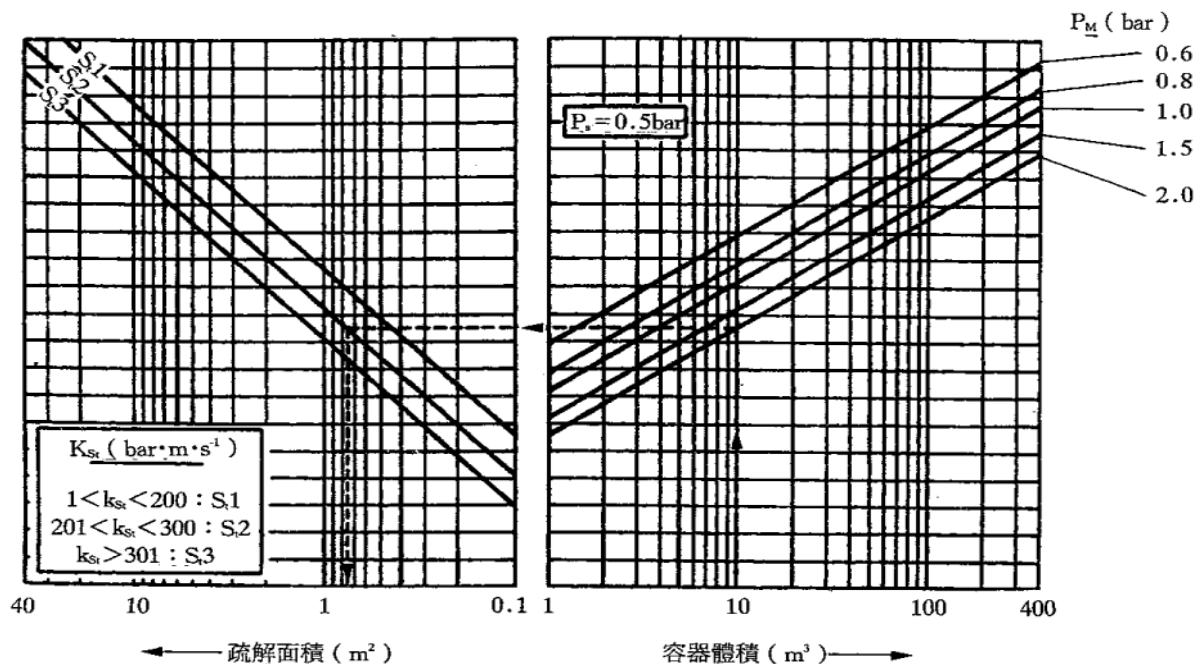


圖2-15 設定壓為0.5巴 (bar) 時，疏解面積、容器體積與塵粒爆炸常數的相互關係圖 [3]

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High Strength Enclosures : Kg(bar-m/sec) for Gas :

MATERIAL	Pmax (bar)	Kg bar-m/sec
ACETOPHENONE	7.6	109
ACETYLENE	10.6	1415
AMMONIA	5.4	10
B-NAPHOL	4.4	36
BUTANE	8.0	92
CARBON DISULFIDE	6.4	105
DIETHYL ETHER	8.1	115
DIMETHYL FORMAMIDE	8.4	78
DIMETHYL SULFOXIDE	7.3	112
ETHANE	7.8	106
ETHYL ALCOHOL	7.0	78
ETHYL-BENZENE	7.4	96
HYDROGEN	6.8	550
HYDROGEN SULFIDE	7.4	45
ISOPROPANOL	7.8	83
METHANE	7.1	55
METHANOL	7.5	75
METHYLENE CHLORIDE	5.0	5
METHYL NITRITE	11.4	111
NEOPENTANE	7.8	60
OCTANOL	6.7	95
OCTYL CHLORIDE	8.0	116
PENTANE	7.8	104
PROPANE	7.9	100
TOLUENE	7.8	94

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High Strength Enclosures : Kst(bar-m/sec) for Dust :

MATERIAL	Pmax,bg	Kst	Haz.Class
ADIPIC ACID	8	97	1
ALUMINUM	12.4	415	3
ANTHRAQUINONE	10.6	364	3
ASCORBIC ACID	9	111	1
BETA CYCLODEXTRIN	7.7	49	1
BRONZE	4.1	31	1
CALCIUM ACETATE	6.5	21	1
CALCIUM STEARATE	9.1	132	1
CARBOXYMETHYLCELLULOSE	9.2	136	1
CELLULOSE	9.7	229	2
CELLULOSE PULP	9.9	62	1
CHARCOAL, ACTIVATED	7.7	44	1
CHARCOAL, WOOD	9	10	1
COAL, BITUMINOUS	9.2	129	1
COKE, PETROLEUM	7.6	47	1
CORK	9.6	202	2
CORN	9.4	75	1
DETERGENT	10	75	1
DEXTRIN	8.8	106	1
EGG WHITE	8.3	38	1
EPOXY RESIN	7.9	129	1
IRON CARBONYL	6.1	111	1
LACTOSE	7.7	81	1
LAMPBLACK	8.4	121	1
LEAD STEARATE	9.2	152	1
LIGNITE	10	151	1
MAGNESIUM	17.5	508	3
MELAMINE RESIN	10.2	110	1

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High Strength Enclosures : Kst(bar-m/sec) for Dust :

MELAMINE, MOLDED (CELLULOSE)	10	127	1
MELAMINE, MOLDED (WOOD)	7.5	41	1
METHYLCELLULOSE	9.5	134	1
MILK, NON-FAT, DRY	8.8	125	1
MILK, POWDERED	5.8	28	1
PAPER DUST	9	200	2
PARAFORMALDEHYDE	9.9	178	1
PEAT, 15% WATER	60	157	1
PEAT, 22% WATER	125	69	1
PHENOLIC RESIN	9.3	129	1
POLY-ACRYLAMIDE	5.9	12	1
POLY-ACRYLONITRILE	8.5	121	1
POLY-ETHYLENE	8	156	1
POLY-METHYL ACRYLATE	9.4	269	2
POLY-METHYL ACRYLATE POLYMER	10.1	202	2
POLY-PROPYLENE	8.4	101	1
POLY-VINYL ACETATE	8.6	119	1
POLY-VINYL ALCOHOL	8.9	128	1
POLY-VINYL BUTYRAL	8.9	147	1
POLY-VINYL CHLORIDE	7.6	46	1
SODIUM ASCORBATE	8.4	119	1
SODIUM STEARATE	8.8	123	1
SOOT, PINE	7.9	26	1
SOY FLOUR	9.2	110	1
STARCH, CORN	10.3	202	2
STARCH, RICE	9.2	101	1
STARCH, WHEAT	9.9	115	1
SUGAR	8.5	138	1
SUGAR, BEET	8.2	59	1
SUGAR, MILK	8.3	82	1
SULFUR	6.8	151	1
TAPIOCA	9.4	62	1
TERPENE PHENOL RESIN	8.7	143	1
UREA FORMALDEHYDE CELLULOSE	10.2	136	1
VINYL ACETYLENE COPOLYMER	8.3	98	1
VINYL ACETYLENE EMULSION	8.2	95	1
WHEY	9.8	140	1
WOOD FLOUR	10.5	205	2
ZINC	7.3	176	1

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以一個蒸餾塔系統為例，計算安全閥排放量，相關資料與流程圖如下：

Protected Equipment Name and Condition

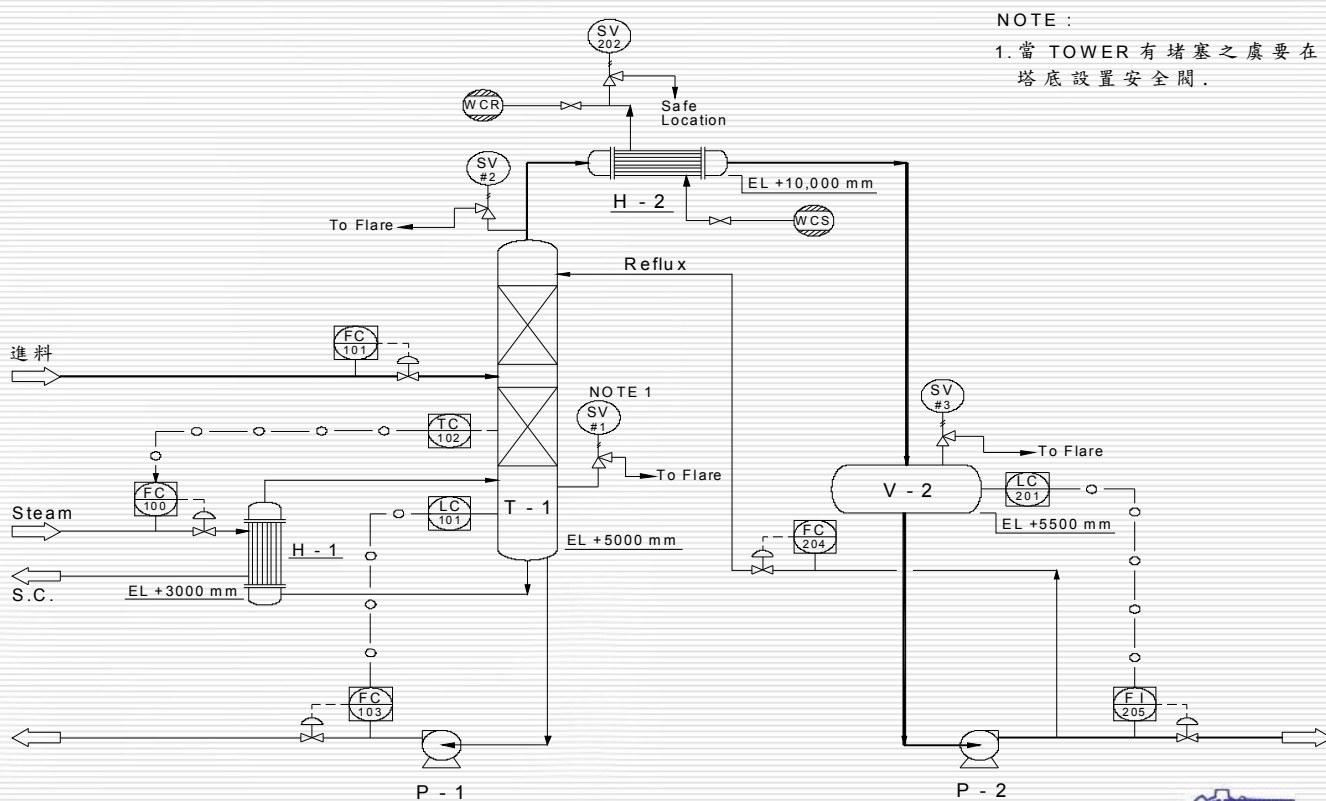
Equip. No.	No.	Head Type (Ellipse or Hemisphere)	Design Press. kg/cm ² G	Design Temp. °C	Diameter mm	Height or Length		Install. (V/H)	BTL EL+	Thickness of insulation mm
						Tube Length mm	Channel Length			
H-1	1	Ellipse	2	135	600	3000	600	V	3000	150
T-1	1	Ellipse	2	135	800	1300		V	5000	150
V-2	1	Ellipse	2	135	1500	2500		H	5500	150

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安全閥排放計算之流程圖



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1. Heat exchanger		H-1	
BTL El.	mm	3000	
Shell diameter,	mm	600	
Tube length,	mm	3,000	
Channel Length	mm	600	
Heat Exchanger No.		1	
Protected Side		T	Tube Side
Wetted area,	m ²	1.91	$(\pi D L + 2 * 1.084 D^2) / 1000000$
Thickness of insul., d	mm	100	
Environment factor,	F	0.3	
2. Vessel			
BTL El.	mm	5500	
Diameter	mm	1,500	
Height	mm	2,500	
Head Type (1. 2:1 Ellip. ; 2. Hemisphere)		1	
Installation (Ver.=1 ; Hor.=2)		2	
Wetted area,	m ²	12.49	Vertical : $(\pi D L + 2 * 1.084 D^2) / 1000000 * 0.75$
Thickness of insul., d	mm		
Environment factor,	F	1	

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3. Tower			
BTL El.	mm	5,000	
BTM diameter,	mm	800	
BTM height,	mm	1,400	NNL of tower bottom
Tower Wetted area,	m ²	4.21	Vertical : $(\pi D L + 2 * 1.084 D^2) / 1000000$
Thickness of insul., d	mm	100	
Environment factor,	F	0.3	

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1. External Fire Case			
Set Pressure	kg/cm ² G	2.0	Set pressure = Design Pressure
Relieving Pressure	kg/cm ² G	2.4	(1+21%)*Set Pressure
Relieving temperature, °C		106.5	
Molecular weight		31.9	
Latent heat,	kcal/kg	250.0	
Heat Capacity	kcal/kg-°C	0.31	
Fluid State (Gas / Vapor / Steam / Liquid)		V	
For API Equation			
$Q=C_f * F * A^{0.82}$	kcal/hr		$C_f=37,160$
Heat Exchanger	Kcal/hr	18963	$Q=C_f * F * A^{0.82}$
	kg/h	76	= Q / Latent Heat
Vessel	Kcal/hr	464,286	
	kg/h	1,857	
Tower	Kcal/hr	46,959	
	kg/h	188	
Overall	kg/h	2,121	

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2. COOLING WATER FAILURE			
$Q=U*A*(ts-tb)$			$T_s=151.4^{\circ}\text{C}$; $T_t=135.4^{\circ}\text{C}$ (bottom temperature at relief pressure)
Set Press.,	kg/cm ² G	2.0	
Relieving Press.,	kg/cm ² G	2.2	(1+10%) Set Pressure
Relieving temperature, °C		120.1	
Molecular weight		38.9	
Latent heat,	kcal/kg	250	
Fluid State (Gas / Vapor / Steam / Liquid)		V	
Steam temperature, °C		151.4	
Reboiler overall coeff., U	kcal/m ² -h-°C	516.7	
Reboiler surface area, A _r	m ²	47.5	
Heat input,	kcal/h	392,692	
Capacity,	kg/h	1,571	Heat input / Latent Heat, (Normal Rate = 3114 kg/hr)

- ◆ 以本例而言，是以火災所引起之排放量(2121 kg/hr)最大，故安全閥以火災為設計Case。
- ◆ 至於Chemical Reaction與Control Valve Closed Outlet & Failure在本例則無須考慮。

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No.		SV-202
Service		H-2
Fluid		Water
Protection Side		Shell
U, Overall Coefficient (For Thermal Expansion)	kcal/m ² -h-°C	200.0
A, Surface (From HX spec.)	m ²	73.1
a, Coeff. 1 (From Chemical Properties Handbook)		5.7160E-04
m, Coeff. 2 (From Chemical Properties Handbook)		-0.7143
T _C , Critical Temperature	K	647.13
P ₁ , Operating Pressure	kg/cm ² G	3.50
T ₁ , Operating Temperature (Cold Side)	°C	33.0
P _{ext} , External Pressure	kg/cm ² G	0.99
T _{ext} , External Temperature (Hot Side)	°C	66.6
P _d , Design Pressure (or Set Pressure)	kg/cm ² G	5.00
ρ _{liq} , density of Liquid @ Operating Temperature	kg/m ³	995.1
C _p , Heat Capacity of Liquid @ Operating Temperature	kcal/kg-°C	0.999

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β _{liq} , Thermal Expansion Coefficient ($\beta_{liq} = a \cdot (1 - T/T_C)^m$)	1/°C	9.03E-04
P ₂ , Relieving Pressure ($P_2 = (1 + 10\%) \cdot \text{Set Pressure}$)	kg/cm ² G	5.5
ΔT = T _{ext} - T ₁	°C	33.6
Q _v , Volumetric Expansion Rate ($Q_v = \beta_{liq} \cdot \rho_{liq} \cdot C_p \cdot UA \Delta T$)	m ³ /h	0.446
m _c , Mass Expansion Rate ($m_c = Q_v / \rho_{liq}$)	kg/h	444

以本例而言，H-2因為冷卻水關閉後，因熱膨脹所引起之排放量為 444 kg/hr，故安全閥以此為設計Case。

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由於常壓儲槽(設計壓力 ≤ 15 psig)之Emergency Venting，一般是根據API 2000相關規定來計算，要考慮之情況為 **Internal Heating Coil破裂**或**外部火災**，其**吸熱量**請參考以下之公式：

Wetted Surface Area (m ²)	Design Pressure (barg)	Heat Input (Watts)
<18.6	≤ 1.034	$Q = 63,150A$
≥ 18.6 and <93	≤ 1.034	$Q = 224,200A^{0.566}$
≥ 93 and <260	≤ 1.034	$Q = 630,400A^{0.338}$
≥ 260	between 0.07 and 1.034	$Q = 43,200A^{0.82}$
≥ 260	≤ 0.07	$Q = 4,129,700$

上表中吸熱量的公式隨Wetted Area大小而異，其基本的概念是當儲槽愈大，單位液體接觸面積的吸熱量愈低。對於wetted area ≥ 260 m²，設計壓力 ≤ 0.07 barg的儲槽，由於整個儲槽不致全部陷於火場中，但其未與液體接觸的部分卻可能在最大的氣化量未達到前，金屬的部分已經因為高溫失去其強度，因此，火災造成其內容物氣化之最大吸熱量僅考慮約4100 kW；至於設計壓力0.07~1.034barg的儲槽，由於其內容物的儲存條件通常接近於其沸點，因此火災使其內容物達沸點的時間應不需很長，因此仍建議用API 521中計算壓力大於15 psig的設備在火災中的吸熱量公式算。

火災緊急排放量計算：

$$Nm^3/h = 881.55 \times \frac{QF}{L} \times \left(\frac{T}{M} \right)^{0.5}$$

- Nm³/h= venting requirement, in normal cubic meters per hour of air.
 Q = heat input from fire exposure, in Watts.
 F = environmental factor from Table 4B. Credit may be taken for only one environmental factor,
 L = latent heat of vaporization of the stored liquid at the relieving pressure and temperature, in joules/kilogram (J/kg),
 T = temperature of the relieving vapor, in degrees Kelvin. It is normally assumed that the temperature of the relieving vapor corresponds to the boiling point of the stored fluid at the relieving pressure,
 M = molecular weight of the vapor.

計算例：

有一個ID為6m且高度為12m之儲槽，該儲槽無保溫，設計壓力250mmAq.，內容物為醋酸（latent heat= 399,036 J/kg），NBP = 118.1°C，MW = 60.05，請計算火災時之緊急排放量為多少Nm³/hr？

1. 首先判斷高度是否高於30ft (9.144m)，按照API 2000之規定，高於30ft時以 30ft來計算，不足30ft時，以實際高度計算，以本例而言是以9.144m來計算。
2. Wetting Area (A) = $3.1416 \times 6 \times 9.144 = 172 \text{ m}^2$
3. 火災吸熱量 (Q) = $630,400 \times 172^{0.338} = 3,593,619 \text{ Watts}$
4. 環境因素 (F) = 1（桶槽無保溫）
5. 緊急排放量
= $881.55 \times (3,593,619 / 399,036) \times 1 \left[(118.1 + 273.15) / 60.05 \right]^{0.5}$
= 20,265 NM³/hr

- ◆ **ISA-20 Specification Form**（[附件B](#)），針對各個欄位都有很詳細之說明。
- ◆ **API 520**：PRV及Rupture Disk規格書樣本（[附件C](#)）
- ◆ 背壓值會影響廠商報價，需正確計算，必要時透過與Vendor之確認後，再做適當之修正。
- ◆ 利用廠商提供之免費軟體，可以先行計算進出口管嘴之大小，但不同廠商有可能不同，因此最終仍須以得標廠商報價之尺寸為準。

Fig.5 - Typical Pressure Relief Valve Without a Stop

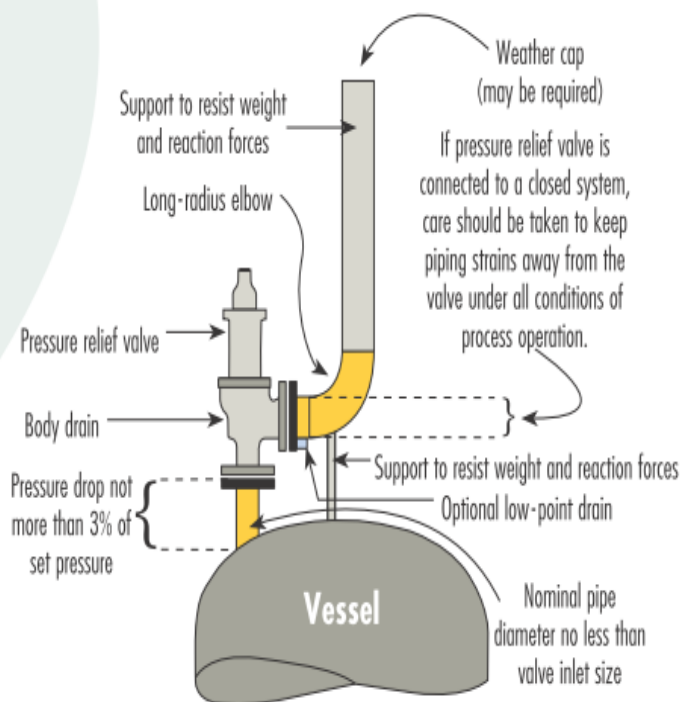
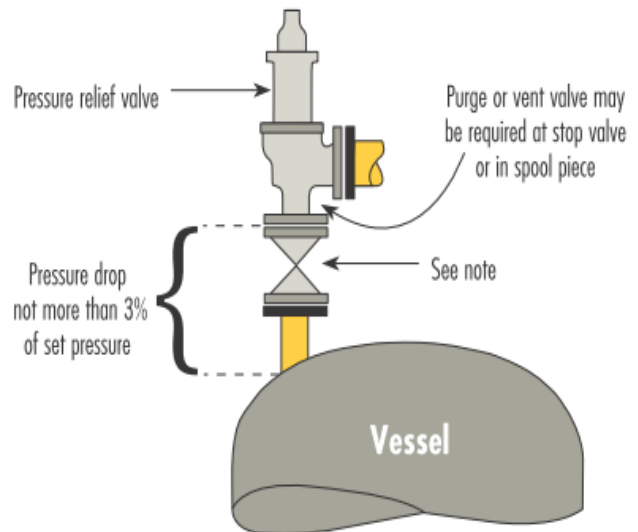


Fig. 6 - Typical Pressure Relief Valve With a Stop Valve



NOTE: The stop valve must have a full port area greater than or equal to the inlet size of the pressure relief valve. The stop valve should be used only as permitted by the applicable codes.

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Fig. 7 - Typical Pressure Relief Valve Mounted on a Process

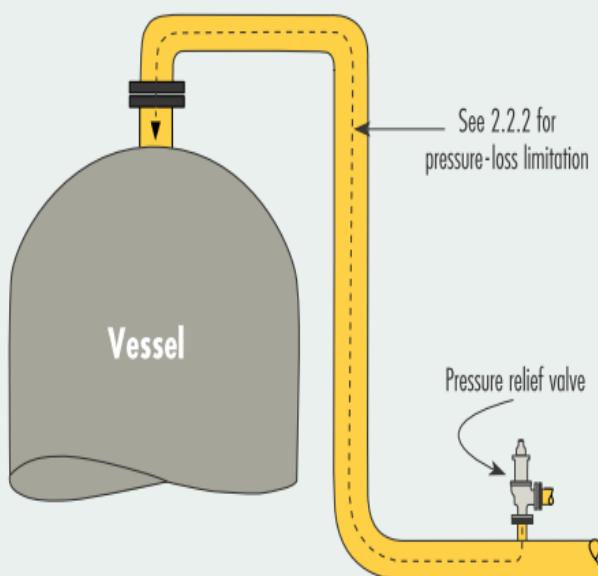
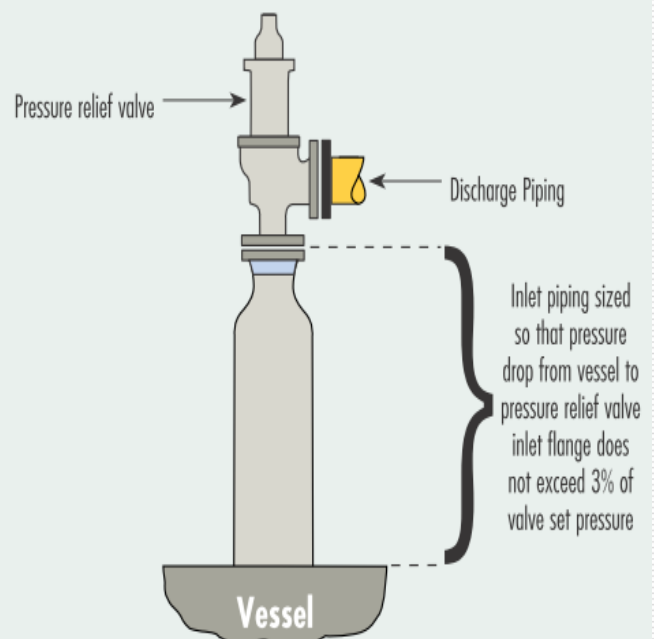


Fig. 8 - Typical Pressure Relief Valve Mounted on a Long Inlet



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Fig. 9 - Typical Pilot Operated Pressure Relief Valve Installation

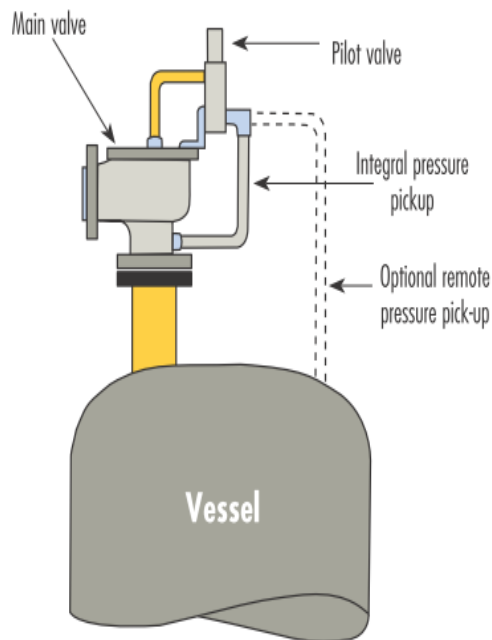
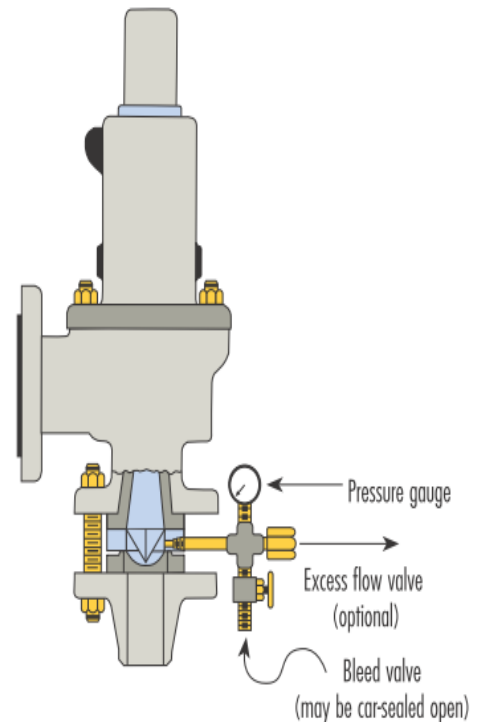


Fig. 10 - Typical Rupture Disk Assembly Installed in Combination with a Pressure Relief Valve



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Fig. 11 - Recommended Typical Installation to Avoid Process Laterals Connected to Pressure Relief Valve Inlet

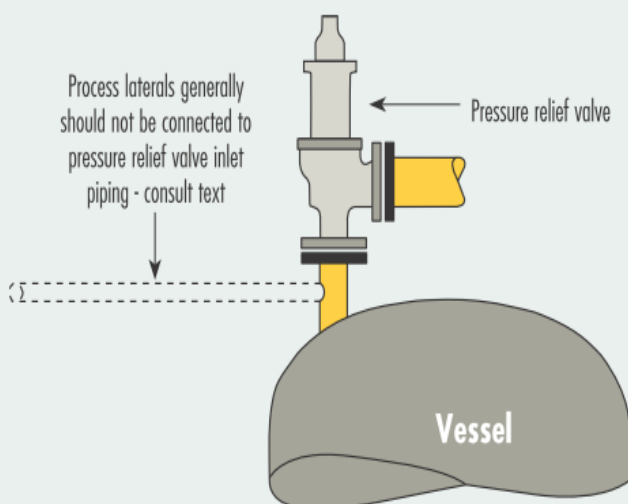
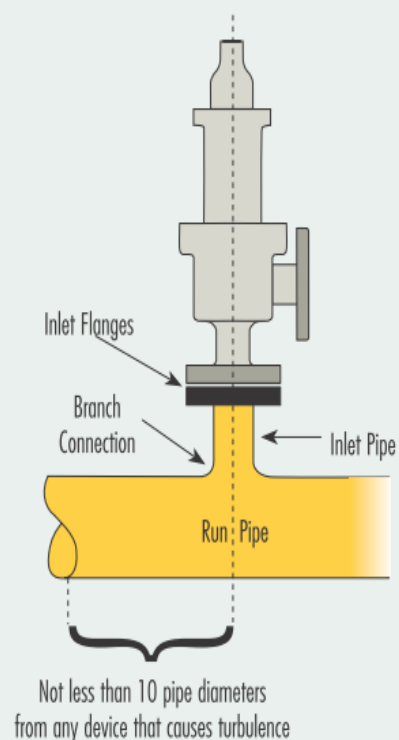


Fig. 12 - Typical Installation Avoiding Excessive Turbulence at Pressure Relief Valve Inlet



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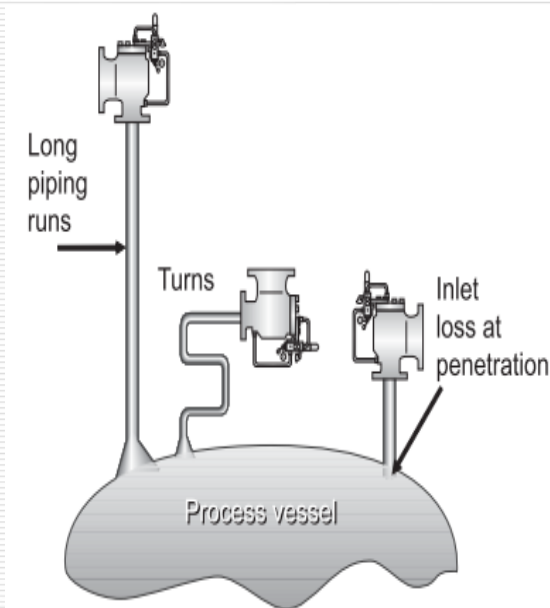


FIGURE 6.4

Causes for excessive inlet pressure losses

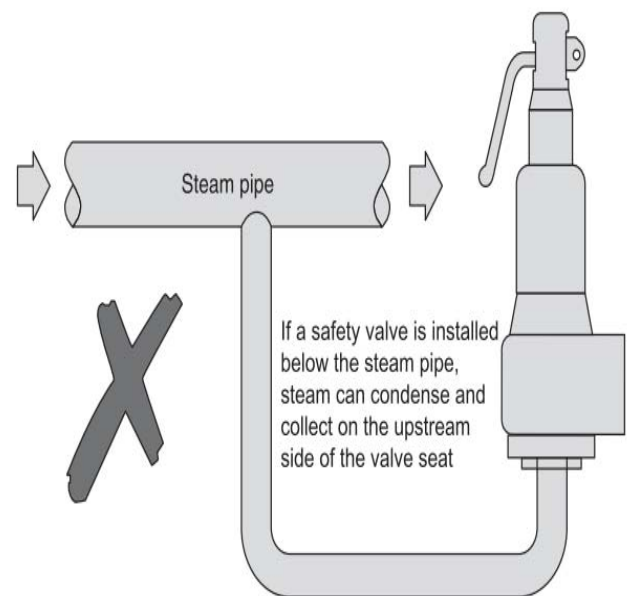


FIGURE 6.19

Avoid possible formation of pockets in the inlet piping

In general, manifolds must be sized so that in the worst case (i.e. when all the manifold valves are discharging at the same time), the downstream piping is large enough to cope without generating unacceptable levels of backpressure. The volume of the manifold should ideally be increased as each valve outlet enters it, and these connections should ideally enter the manifold at an angle no greater than 45° to the direction of flow, as shown in Figure 6.20. The manifold must also be properly secured, supported and drained where necessary.

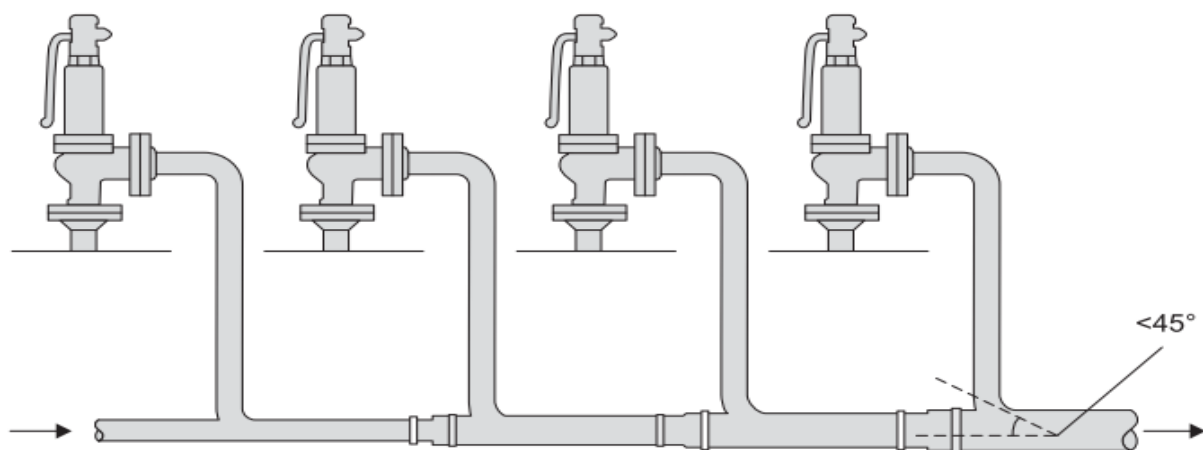
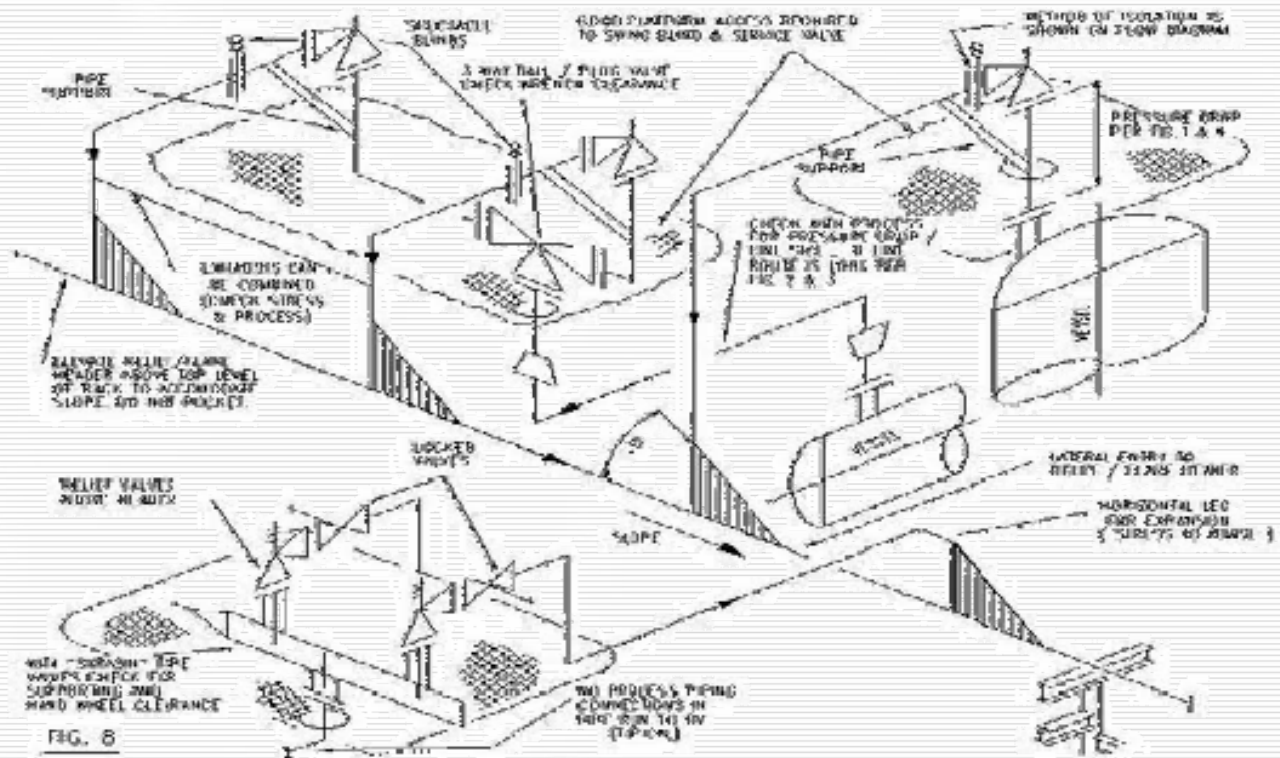


FIGURE 6.20

Manifold discharge system



RECOMMENDED TYPICAL RELIEF VALVE INSTALLATION

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HIPS系統簡介

- 在API 521 Annex E (informative)有特別提到，除了傳統機械式安全閥之保護設施以避免系統過壓外，另一個新的趨勢為採用High-Integrity Protection System (HIPS)儀表系統來增加系統之安全與可靠性。
- 適當規劃的HIPS系統，可以提供等同或優於傳統之機械釋壓裝置。
- 在某些高風險系統，HIPS系統搭配傳統之機械釋壓裝置，可以達到高可靠度之保護等級。

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E.2.1 Elements of a HIPS

A HIPS includes field instruments (e.g. sensors), logic solving devices (e.g. Safety System Logic Solver, relays, etc.), final control elements, power supply and inspection, testing, and maintenance procedures. The boundaries of a HIPS incorporate all aspects from the sensor to the final element.

E.2.2 Application of HIPS

There are five principal uses of a HIPS:

- a) to eliminate a particular overpressure scenario from the design basis;
- b) to eliminate the need for a particular relief device;
- c) to provide system overpressure protection where a relief device is ineffective;
- d) to reduce the probability that several relief devices will have to operate simultaneously, thereby allowing for a reduction in the size of the disposal system;
- e) to reduce the demand rate on a relief device consequently reducing the risk.

謝謝指教！